

The Expansion and Dynamic Equilibrium Effects of Institutional Landlords*

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Abstract

This paper studies how dynamically formed cost efficiencies from scope and density drive institutional landlords' expansion and, in turn, alter welfare distribution across heterogeneous households in the single-family home markets. Institutional landlords convert owner-occupied homes into large, spatially clustered rental portfolios. They constrain the supply of homeownership while expand rental opportunities, leading households to reoptimize between buying and renting, as buyers may face higher prices while renters may benefit from expanded choice sets. We build a dynamic equilibrium model of landlord investment with three key features: (i) endogenous housing supply evolution determined by oligopolistic landlords' investment, (ii) endogenous landlord costs varying with portfolio size and density, and (iii) households' substitution within and across buying and renting in an integrated choice set. We estimate the model using firm-property level data from 2013-2022 in the Atlanta metropolitan area. We find that institutional landlords' expansion achieved 60.03% maintenance costs reduction from economies of scope and density. Household total welfare increased with varying effects across renters and buyers. The majority of renters gained from expanded rental supply, while a small fraction of renters, together with most buyers, lost from diminished access to affordable homeownership. These findings have significant policy implications on regulating institutional landlords' expansion in the single-family home market.

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1 Introduction

The rise of institutional landlords over the past decade has reshaped the single-family housing supply in Sun Belt metro areas. These firms convert owner-occupied homes into large, spatially clustered rental portfolios, constraining the supply of homeownership while expanding rental opportunities. Supply reallocation leads households to reoptimize between buying and renting: buyers may face higher prices, while renters may benefit from expanded choice sets. In this paper we ask how dynamically formed cost efficiencies from scope and density drive institutional landlords’ expansion in competition and, in turn, alter welfare distribution across heterogeneous households.

To answer this question, we analyze linked datasets of property transactions, rental listings, resident demographics, and institutional ownership in the Atlanta metropolitan area. We document three key patterns in institutional landlords’ expansion. First, institutional landlords have expanded fast by actively acquiring single-family homes and converting them into rentals. From 2013–2022, the top six institutional landlords acquired over 50,000 single-family homes, or 3% of the total housing stock. Second, these landlords tend to form spatially clustered portfolios with location priorities differentiated from competitors. By 2022, 65.8% of their rental units were located in areas with more than 25 units within a one-mile radius managed by the same landlord. Third, institutional landlords target their acquisitions toward lower-priced segments, with purchase activities peaking among 20th–30th percentiles of sale prices among all homes for sale. Finally, we find correlational effects of institutional landlords’ expansion on the local housing market. In the ownership segment, their expansion correlates with higher sale prices, increased transaction volumes, and a reduction in individual home purchases. In the rental segment, their investment coincides with expanded total rental supply with no clear price effects¹.

Motivated by these patterns, we develop a dynamic equilibrium model of landlord investment with a micro-founded static housing market equilibrium solved in each period. Both components are essential for understanding the economic drivers of institutional landlords’ long-term expansion, and for determining the welfare implications of their expansion. In each period, institutional landlords engage in oligopolistic competition: they adjust their rental portfolios by choosing how many and which types of homes to buy or sell, endogenizing the marginal effects on equilibrium prices, rents, and household choices in their optimal decision making. Unlike small landlords, each institutional landlord faces endogenous per-unit costs of maintenance that vary with portfolio size and local density. Optimal investment balances an intertemporal trade-off between rising acquisition costs today and expected increases in future profit streams. The latter depends on the extent to which portfolio growth-driven cost reductions outweigh the downward pressure on rents under rental supply expansion.

¹We find very small (in scale) yet statistically significant negative effects on rental prices for properties run by local small landlords, and very small (in scale) yet statistically significant gap in rental prices between small landlords’ properties (lower) and institutional landlords’ properties (higher).

In each period, we micro-found housing demand with discrete housing choices made by heterogeneous households among a unified set of differentiated options across ownership and rental segments. The assumption of endogenous demand for ownership and rental is relevant to rationalizing landlords’ investment target and to welfare implications. It captures all types of substitutions within and across buying and renting when households re-optimize upon institutional landlords’ expansion. Finally, the housing market equilibrium is completed with supply of homeownership interacting with prices, and supply of small landlord rentals aggregated from landlords’ entry-exit decisions.

We estimate the model using data on institutional landlords’ investment, household housing choices, and incumbent supply decisions in Atlanta single-family home market from 2013-2021. Parameters of interest include housing demand, homeownership supply, small-landlord rental supply, and institutional landlords’ investment costs and maintenance costs. Estimation proceeds in two steps. First, we recover the static demand parameters using Generalized Method of Moments (GMM) estimator, leveraging the exogenous part of cross-market variations in housing supply structure as an instrument for housing prices and to identify the nesting parameter. We leverage cross-market variations in housing population demographics to instrument for endogenous prices in the homeownership supply and small landlord rental supply, to identify static supply parameters. In the second step, we follow the two-step estimation framework and adapt it to the multi-product firms’ continuous choice case. We first fit the empirical policy function of investment using Neural Network to improve the flexibility of functional forms. Using estimated policy functions and state transition parameters, we forward-simulate equilibrium paths of the dynamic investment game and approximate the expected value functions. Forward-simulations for states on firms’ equilibrium paths and deviated paths with one-step difference from their choice of action in data. With approximated value function gradients, we convert firms’ optimality conditions in each dimension of the multi-product portfolio investment into a joint likelihood function and recover cost parameters using maximum-likelihood estimation (MLE).

Our estimates imply that middle-income households view the purchase and rental of lower-priced homes as close substitutes. With ownership prices bid up and rental supply expanded by institutional landlords, a larger share of these households rent. Cost estimates reveal substantial economies of scope and density, driving landlords’ spatial clustering in expansion. Our estimates also indicate a low supply elasticity of homes for purchase, inducing landlords to spread their investment across periods and locations to mitigate the upward pressure on acquisition costs.

Counterfactual analysis reveals mixed welfare effects of institutional landlords’ expansion. In primary markets, renter shares increase by 1.58%. Over the years, the welfare gain is \$50.17 per renter year, with 84.04% benefiting from the expansion of rental supply and 15.96% losing due to diminished access to their preferred affordable ownership options. Remaining buyers incur a net loss of \$31.35 per buyer year from inflated ownership prices. Expansion-driven scope and density

reduce costs by 60.03%. We conclude that endogenous cost efficiencies play a central role in driving housing supply reallocation, a mechanism amplified when investors are forward-looking.

Related Literature. This paper is the first to build and estimate a dynamic equilibrium model to analyze the expansion and welfare impacts of institutional landlords in the single family home market. Reduced-form studies have documented causal effects of the presence of institutional landlords on housing market outcomes (Garriga et al., 2020; Gurun et al., 2022) and neighborhood gentrification (Austin, 2022), leveraging local exposure to landlord merger events as exogenous treatment. There is a growing strand of papers using structural methods to study the equilibrium effects of institutional landlords, emphasizing different channels of their impact. Barbieri and Dobbel (2025) study the potential market power of institutional landlords in pricing, as a static equilibrium outcome conditional on (i) exogenous housing supply structure and fixed institutional landlords’ holdings, and (ii) exogenous total demand allocated to the rental market and the ownership market. Our paper abstracts away from the short-term pricing margin, instead, investigates the long-term investment, highlighting (i) endogenous evolution of housing supply shaped by institutional landlords’ dynamic investment choices, and (ii) flexible demand substitution within and across renting and buying. Existing research on institutional landlords’ investment treats costs as an exogenous static variable (Coven, 2025), as a time-series trend declining over time (Gorback et al., 2025), or as a static choice made by a single global investor (Chang, 2024). We instead, explicitly model the formation of landlord cost efficiency as a series of strategic choices made in interactions with market conditions and competition landscapes. Given the inelastic supply of homeownership and other frictions, housing markets evolve gradually with rich intertemporal trade-offs in decision making, which are better captured in a dynamic framework.

More broadly, this paper adds to the existing literature of industrial organization in housing markets. Existing literature have brought demand estimation methods to various contexts with housing or neighborhood choices (Bayer et al., 2007, 2016; Calder-Wang, 2021; Almagro et al., 2024; Calder-Wang and Kim, 2024; Cook et al., 2025), with most applications focusing on either the rental or the home purchase market. This paper proposes an integrated market with housing supply in two segments, rental and ownership, modeled using nested logit, providing a stylized yet effective way to capture substitution across tenure types, which is relevant to welfare implications when households re-optimize their tenure choices². In addition, this paper contributes to a growing literature on endogenous housing supply structure. Examples include Diamond et al. (2019) on rental supply responses upon expansion of rent control in San Francisco; Farronato and Fradkin (2022) and Calder-Wang (2021) on the rise of short-term rental supply when Airbnb platform arrives. This paper investigates the long-term housing stock conversion from the owner-occupied segment to the rental segment through firms’ dynamic investment. The framework can be generalized to study a

²Examples include Han et al. (2022) on property transaction tax’s impacts on housing tenure choices, and Greenwald and Guren (2025) on quantifying the level of housing market segmentation by tenure.

broader range of real estate investment problems.

Our findings highlight the role of dynamically-formed economies of scope and density in driving institutional landlords’ expansion in the single-family home market. Existing literature have documented the presence and evolution of economies of scale, scope, and density in retail chains (Jia, 2008; Holmes, 2011; Ellickson et al., 2013), newspaper circulation (Fan, 2013), transportation networks (Chen, 2024), and online platforms’ distribution networks (Houde et al., 2023). We extend the literature to housing markets featuring inelastic supply of homes for purchase in each period. Given this, we provide an analysis on the optimal timing of cost efficiency formation, which is less studied in previous work. In contexts with local resource constraints, fast densification of network is penalized with rising costs in investment, which is more prominent when multiple firms compete for the constrained resources to densify their networks in the same location. To mitigate congestion costs, optimal expansion strategies tend to spread smaller-scale investments over multiple periods and spatially differentiate their investments from competitors.

This paper is the first to allow both endogenous costs and benefits of dynamic investment to be determined through a market. Endogenous investment costs are specifically important in industries featuring scarce inputs, including plantation development (Hsiao, 2025) and oil exploration (Hodgson, 2024). Crowded investment leads to congestion in factor markets, resulting in increases in investment costs. Our framework micro-founds this mechanism, allowing rich elasticities in the factor market to flexibly impact firms’ choice of investment via cost incentives.

Methodologically, this paper builds on the literature on dynamic estimation (Hotz et al., 1994; Bajari et al., 2007; Akerberg et al., 2007) and provides estimation for a continuous-choice dynamic investment game with multi-product oligopolistic firms. We follow the two-step estimation framework proposed by Bajari et al. (2007) (henceforth BBL) and contribute from two perspectives. First, a rich body of empirical literature has explored the BBL estimation method for dynamic agents with discrete choices (Ryan, 2012; Collard-Wexler, 2013; Fowlie et al., 2016; Caoui et al., 2022), but less has been studied about dynamic agents making continuous choices with the exception of Barwick et al. (2025) that analyzes industrial policies’ implications on shipbuilding investment. Recent advances in methods include Gowrisankaran and Schmidt-Dengler (2024), developing computable methods to solve dynamic oligopoly models with many ordered investment choices in a capacity game. This paper contributes the first application to continuous investment problems in spatial competition, where heterogeneous firms optimize by adjusting the densities of their networks over a finite grid of discrete locations. This approach can be generalized to a broader set of problems studying network competition between firms operating in a spatial context. Second, following Sweeting (2013) on radio stations and Bodéré (2023) on pre-schools, we micro-found payoff flows with rich static market structures featuring choices made by heterogeneous consumers in equilibrium. We contribute a fully endogenous payoff flow example with both investment costs and per period returns determined by the static equilibrium.

2 Background and Data

2.1 The Rise of Institutional Landlords in Single-family Home Markets

Institutional participation in the single-family housing market grew sharply in the aftermath of the 2008–2009 financial crisis, when foreclosure waves created large inventories of real-estate-owned (REO) properties available for bulk purchase. The phenomenon was most evident in the Sun Belt metropolitan areas, where the pre-crisis housing bubble had been more pronounced and the correction correspondingly more severe. The opportunity of acquiring discounted properties at scale attracted the earliest wave of institutional investment, as these investors held a relative advantage in access to capital and financing. Later on, as the regional economy gradually recovered, inland Sun Belt markets—including Atlanta, Dallas, and Charlotte—experienced strong job growth that attracted middle-income, working-age households and young families. This migration increased demand for affordable housing and contributed to the stabilization of local housing markets. Given the trends, many institutional investors transitioned from short-term asset purchases to developing long-term single-family rental portfolios, consolidating their holdings into large-scale landlord entities. After ten-year sustained acquisitions, institutional owners held roughly 450,000 rental homes nationwide by 2022, which represent a modest share of the total single-family housing stock but a high prevalence in selected Sun Belt metropolitan areas.

We make two clarifications about the scope of this paper. First, this paper focuses on the long-term expansion of institutional landlords. Questions on the initial rise of these landlords are beyond the scope of this paper, see [Mills et al. \(2019\)](#) and [Hanson \(2022\)](#) for more discussions. Second, to the best of our knowledge, there is no official criterion that defines an institutional landlord. For clarity, in this paper we define institutional landlords in Atlanta as large corporate owners managing rental portfolios of $\geq 1,000$ single-family homes in the area by 2022, along with previously active large corporate landlords that were subsequently acquired or merged into these entities. Figure 1 summarizes the evolution of portfolio sizes for the institutional landlords studied in this paper.

2.2 Institutional landlords’ Rental Business Model

Institutional landlords acquire, renovate, and lease single-family homes. When acquiring properties, they typically pay the lump sum ownership costs with cash, instead of using mortgage financing, which allows transactions to close quickly. Since 2014, institutional landlords have shifted from acquiring foreclosure sale homes to buying homes from the retail markets, where most home sellers are owner-occupiers, followed by local homebuilders and small corporate owners. Many institutional landlords work with local real estate agents and brokerage networks to search for suitable homes listed for sale. By 2022, purchases from the ongoing market have accounted for more than 90% of

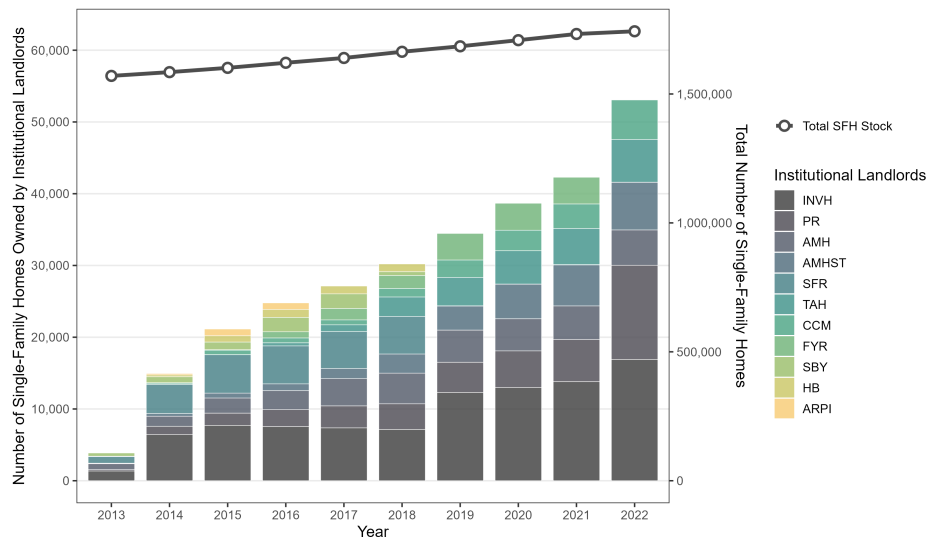


Figure 1: Evolution of Total Housing Stock and Institutional Landlord Holdings in the Single-Family Home Market, Atlanta MSA 2013-2022

their rental stock.

Post acquisition, institutional landlords undertake a period of renovation to convert homes in varying conditions into rental properties ready for leasing. Renovation involves interior and exterior repairs and replacements, painting, landscaping, and compliance with local building and rental codes. Eligible rental homes are then offered to tenants through leasing platforms, tagged as standardized branded rental products. Despite differences in operational procedures across firms, institutional landlords exhibit broadly similar patterns of property management. Most employ centralized administrative systems, such as digitized lease processing, automated rent collection, and property condition monitoring, and implement standardized leasing and maintenance protocols. These practices enable firms to manage geographically dispersed portfolios of single-family homes efficiently and to deliver consistent service quality at scale.

The rapid growth of large institutional landlords in the U.S. single-family housing market has generated significant policy interest, from federal agencies to local jurisdictions. For instance, a 2024 Government Accountability Office report reviews the expansion of institutional ownership in the single-family rental sector and discusses its potential effects on home prices and rents, though it finds the evidence on home-ownership opportunities and tenant welfare inconclusive due to data limits (Office, 2024). Despite this heightened attention, few national-level policies currently aim either to restrain or to encourage institutional participation in the single-family rental market.

2.3 Data Sources

2.3.1 Property Ownership Panel

Our primary data is a comprehensive panel of property ownership history for the universe of single-family homes in Atlanta metropolitan area from 2011-2022. We construct the panel by linking two data sources, CoreLogic property tax data and property deeds, at the property level using unique identifiers. For each property, annual tax data informs the property type, land use code, property characteristics, year of built, property street address, property tax payment, assessed value, and tax payer’s mailing address. For each transaction of property ownership, we observe the sale amount, sale date, multi-property sale indicator, and public information on sellers and buyers. Importantly, we observe corporate sellers’ and buyers’ names and leverage them to track institutional landlords’ participation in ownership transactions.

By merging property transaction record with tax history, we partition a property’s history into periods owned by different owners. For each period, we classify each property’s ownership status into 3 types: $\{OO, SL, IL\}$, short for owner-occupied, small landlord, and institutional landlord, using the following criteria: ³ ⁴

In a given tax year, a property is identified as a rental property, if

1. Mailing address of the tax payer differs from the property address, AND
2. Mailing address is not a mortgage lender’s address

For a rental property, we further identify its institutional ownership using buyer’s information from its most recent transaction. Institutional investors use subsidiaries to buy properties, making it challenging to identify the ultimate ownership of these properties. To address the issue, we assemble public information on corporate property buyers and sellers from rich data sources, including institutional investors’ historical SEC filings, Georgia Corporations Division Business Search, and Open Corporates. Table 11 summarizes the cleaning results. On top of these, a rental property’s ownership can change post mergers between two institutional landlords.⁵ To capture the

³If a property’s indicated status changes under the same owner, we adopt the most frequent tenure-owner status to represent the property’s overall status during the consecutive years within the same ownership period. Common reasons for the change include new homeowners continuing to use their previous residence’s address for tax mailing purposes for the first several years after move-in, until they eventually update it to their current residence.

⁴Under this criteria, properties under flipping status are labeled as either owner-occupied or for-rental, based on whether the flippers use the property’s address for mailing address. We believe the noise introduced by this specification is not going to change the conclusion of this paper for the following reasons: (1) properties under flipping each year account for a relatively small fraction (less than 4%) of Atlanta’s SFH stock; (2) the distribution of flipping activities is not significantly different in single-family houses for-rental versus for owner-occupied.

⁵During the period of study from 2011-2022, we document 5 primary M&As between institutional landlords. American Residential Properties → American Homes 4 Rent (2015), Starwood Waypoint Homes → Invitation Homes (2017), Silver Bay Realty → Tricon American Homes (2017), HavenBrook partners → Front Yard Residential (2018), and Front Yard Residential → Progress Residential (2020).

changes, we refine the ownership panel construction using data from the SDC Platinum Mergers & Acquisitions Database.

We include newly constructed properties in the panel data 1 year before its first-time sale to the market. We label its initial ownership status as *NEW*, before it is sold to a buyer in one of the three types $\{OO, SL, IL\}$.

2.3.2 Rental Listings and Prices

We obtain rental listing prices from the Altos Multiple Listing Service (MLS) data, which represents a sampling from the universe of rentals at a given period. Overall, the Altos data covers approximately 25% of all single-family homes for rental in the Atlanta metropolitan area⁶. We process the rental prices to two levels. For stylized facts generation, we use the property-level rental prices associated with property characteristics⁷. For demand estimation, we use the median of rental prices for all units listed in MLS for a given type to represent the type level rental price. For home types with rental prices missing in some years, we fill with interpolated prices using observed prices from adjacent years.

2.3.3 Demographics

We use the American Community Survey (ACS) Public Use Micro Sample to obtain demographics along with housing choice characteristics for households in the housing population. we define housing population as a subset of local residents who have no homeownership in the area at the beginning of a given year.⁸ These households constitute the aggregate housing demand for the period of interest. To capture yearly compositional changes in the housing population, we use ACS 1-year estimates.⁹ Demographic variables of interest include household income, age, education, number of children, race and ethnicity. Housing choice characteristics include a single-family home indicator, number of bedrooms, housing tenure statuses, and monthly housing costs.

2.3.4 Summary Statistics

Table 1 presents descriptive statistics on the single-family home stock and ownership structure across 20 submarkets in the Atlanta metropolitan area in 2022. Overall, the majority (79.92%)

⁶There are two main reasons for the missing rental price data. First, properties with renewed leases remain in the rental market but are not relisted on the MLS until the current contract expires. Second, many rental units are leased through non-MLS channels, a practice especially common among small local landlords.

⁷If a property is listed for multiple times within a year, we adopt the last observation of its price from the last listing record in a year to represent its price of the year.

⁸We rely on two ACS survey questions to identify a household’s initial homeownership status: (i) within one year migration and (ii) homeownership. A household is in previous year’s housing population if moving into the current house within one year, or, currently renting the house, or both.

⁹ACS 1-year estimates provide annual updates on a subset of Public Use Micro Areas (PUMAs) with populations of 65,000 or more. All PUMAs in Atlanta MSA are included in the ACS 1-year estimates panel.

of single-family homes are under owner-occupancy, followed by small landlords' holdings (16.25%). Institutional landlords' holdings account for a small fraction (3.24%) of the single-family housing stock, with their presence varying greatly across different submarkets. Five county markets (Paulding, Henry, Douglas, Newton, and Fulton South) have $\geq 5\%$ of SFH stock under institutional ownership. Around 90% of institutional landlords' holdings are located in half of the twenty submarkets, led by Gwinnett (18.2%), DeKalb (12.2%), Cobb (11.5%), Henry (10.1%), Paulding (7.74%), Clayton (7.18%), Fulton South (6.16%), Douglas (5.79%), Cherokee (4.36%), Newton (3.90%), and Atlanta City (2.19%). These submarkets are primarily inner-ring suburbs in the Atlanta metropolitan area. Table 2 provides summary statistics on housing population demographics and ACS survey-reported housing costs by tenure choices.

Table 1: Summary of Single-family Home (SFH) Stock, Atlanta MSA 2022

Submarkets	SFH Units (Total)	SFH Types (Physical)	Owner-occupied Shares (%)	Small Landlord Shares (%)	Institutional Landlord Shares (%)	New Construction Shares (%)
Barrow	29,513	17	81.32	14.83	2.19	1.66
Bartow	35,343	25	75.20	22.34	1.85	0.61
Carroll	38,409	21	76.23	21.40	1.32	1.05
Cherokee	84,871	45	81.24	15.59	2.72	0.44
Clayton	82,087	29	72.07	23.08	4.64	0.22
Cobb	207,857	113	86.06	10.81	2.94	0.19
Coweta	51,243	25	82.53	14.77	1.57	1.14
DeKalb	204,634	93	78.22	18.27	3.17	0.34
Douglas	46,666	21	76.28	16.64	6.57	0.51
Fayette	40,543	13	87.27	11.00	0.85	0.89
Forsyth	86,232	24	84.32	13.36	1.02	1.30
Fulton (North)	87,738	57	77.62	21.71	0.33	0.34
Fulton (South)	61,755	25	71.10	22.72	5.29	0.89
Gwinnett	243,592	132	81.16	14.26	3.95	0.63
Henry	79,287	37	80.09	11.71	6.74	1.46
Newton	38,636	13	75.83	18.45	5.35	0.37
Paulding	60,029	20	78.49	14.27	6.83	0.41
Rockdale	29,805	17	77.56	18.37	3.68	0.39
Walton	34,227	16	79.36	18.23	1.34	1.07
Atlanta City	94,521	67	79.85	18.66	1.23	0.26
Total	1,636,988	767	79.92	16.25	3.24	0.59

Notes: We summarize the composition of single-family home stock for 20 submarkets in the Atlanta metropolitan area. Remaining counties in the metropolitan area are excluded from analysis for data quality issues or lack of institutional landlords presence by 2022. We define submarkets using counties except Fulton County, which is split into three submarkets: Fulton (North), Fulton (South), and Atlanta City, consistent with local municipal planning documents and MLS area references.

Table 2: Summary of Demographics and Housing Costs, Atlanta MSA 2022

Submarkets	Mean HH Income (\$)	Mean Age	Bachelor's Share (%)	Mean Children	White non-Hispanic(%)	Black (%)	White Hispanic(%)	Asian (%)	Mean Owner Cost (\$)	Mean Rent (SFH, \$)	Mean Rent (non-SFH, \$)
Barrow	64,981	44.3	24.21	0.96	67.07	10.26	13.10	1.11	1,592	1,150	737
Bartow	81,710	46.5	23.27	1.01	60.33	19.53	14.83	0.00	1,482	921	960
Carroll	64,829	42.2	30.02	0.73	55.29	24.82	9.18	0.47	1,279	911	923
Cherokee	89,180	45.5	48.73	0.90	63.14	7.92	19.74	1.75	1,875	1,524	1,481
Clayton	62,151	44.4	23.85	0.89	4.85	78.81	10.55	1.07	1,483	1,157	1,136
Cobb	85,045	41.8	49.27	0.61	33.29	40.93	13.57	3.06	2,028	1,493	1,422
Coweta	77,949	47.1	39.82	0.77	45.61	35.37	12.14	0.00	2,361	1,216	1,076
DeKalb	78,961	43.7	48.31	0.64	23.10	52.25	10.66	5.51	1,867	1,517	1,416
Douglas	78,727	41.7	36.14	0.93	19.37	64.07	8.90	0.25	1,425	1,293	1,274
Fayette	90,330	49.6	62.61	1.21	39.81	34.14	14.87	2.38	1,987	1,712	1,670
Forsyth	122,578	47.6	64.86	1.01	58.73	9.73	6.51	20.75	2,167	1,682	1,462
Fulton (North)	116,158	43.9	71.31	0.50	41.71	17.60	16.15	13.79	2,849	2,068	1,709
Fulton (South)	62,522	45.6	30.42	0.91	5.36	80.36	5.81	0.13	1,472	1,285	1,090
Gwinnett	79,379	42.6	38.37	0.95	19.08	41.40	24.84	8.09	1,956	1,583	1,375
Henry	79,126	43.7	33.79	1.12	17.39	64.97	8.82	1.11	1,766	1,383	1,311
Newton	76,640	46.5	27.90	0.92	26.88	55.71	8.35	1.93	1,692	1,203	1,131
Paulding	76,007	43.5	25.70	0.93	40.93	40.33	4.10	0.99	1,455	1,171	1,158
Rockdale	76,756	46.5	27.96	0.92	26.91	55.66	8.35	1.95	1,691	1,203	1,131
Walton	73,525	47.0	19.37	0.91	57.19	26.28	3.60	0.84	1,590	1,169	779
Atlanta City	83,145	41.3	55.54	0.31	30.87	49.26	5.39	5.40	2,812	1,424	1,501
Total	83,913	44.8	39.76	0.89	37.94	39.31	11.85	3.51	1,855	1,356	1,252

Notes: We summarize demographics for households who enter housing market with no homeownership in 2022. Incumbent homeowners are excluded from the summary. We report monthly housing costs by tenure types and exclude utility payments from calculation. Data Source: American Community Survey Public Use Microdata Sample (PUMS).

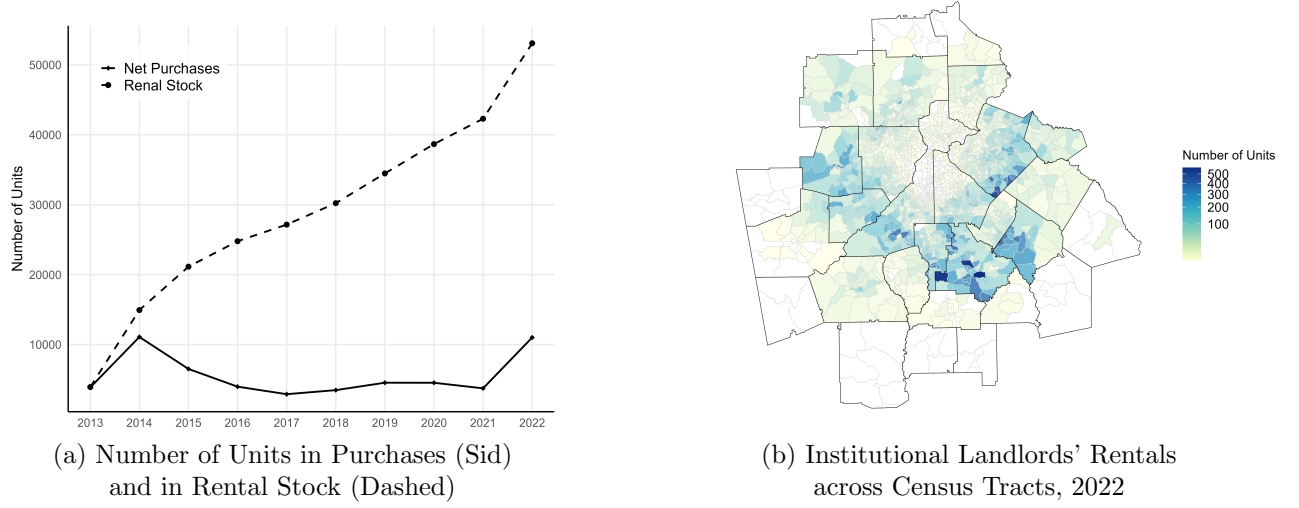
3 Stylized Facts

This section presents stylized facts on institutional landlords' expansion that motivate our modeling choices.

Fact 1: Institutional landlords have grown rapidly, converting homes for purchase into rentals throughout the Atlanta metro area. Figure 2(a) show that institutional landlords grow their rental stock from around 4,000 to 53,000 from 2013-2022, through sustained conversion of homes for purchase into rentals. They participate in 3-11% of year-round home purchases with notable fluctuations across years. By 2022, they account for 15% of the total rental supply in the Atlanta primary single-family home market. Figure 2(b) presents the geographical distribution of institutional rentals by 2022. 512 (or 41.4%) census tracts in Atlanta MSA have more than 50 rental homes owned by institutional landlords. This motivates our modeling choice of a dynamic process in which institutional landlords keeps adjusting the size and composition of their portfolios, through repeated interactions with both ownership and rental segments in the local housing market.

Fact 2: Institutional landlords spatially cluster their investments and differentiate from competitors in target location choices. Looking into the geographical distribution of institutional landlords' rental portfolios, we document a common tendency among landlords to spatially cluster their investment. Table 3 reveals portfolio densification from 2013-2022. A rental home's local density equals 25 if its landlord operates 25 properties within 1 mile from the home of interest. Under this specification, by 2022, 65.8% of institutional landlords' rentals are managed

Figure 2: Institutional Landlords' Purchases and Rental Portfolio Growth, Atlanta MSA 2013-2022



in property networks with greater than 25 local density, marking a drastic increase from 2.7% in 2013. See heatmap plots of property network density of each institutional landlord in Figure 8.

Table 3: Share of Institutional Landlord Rentals by Local Density Ranges (%), Atlanta MSA

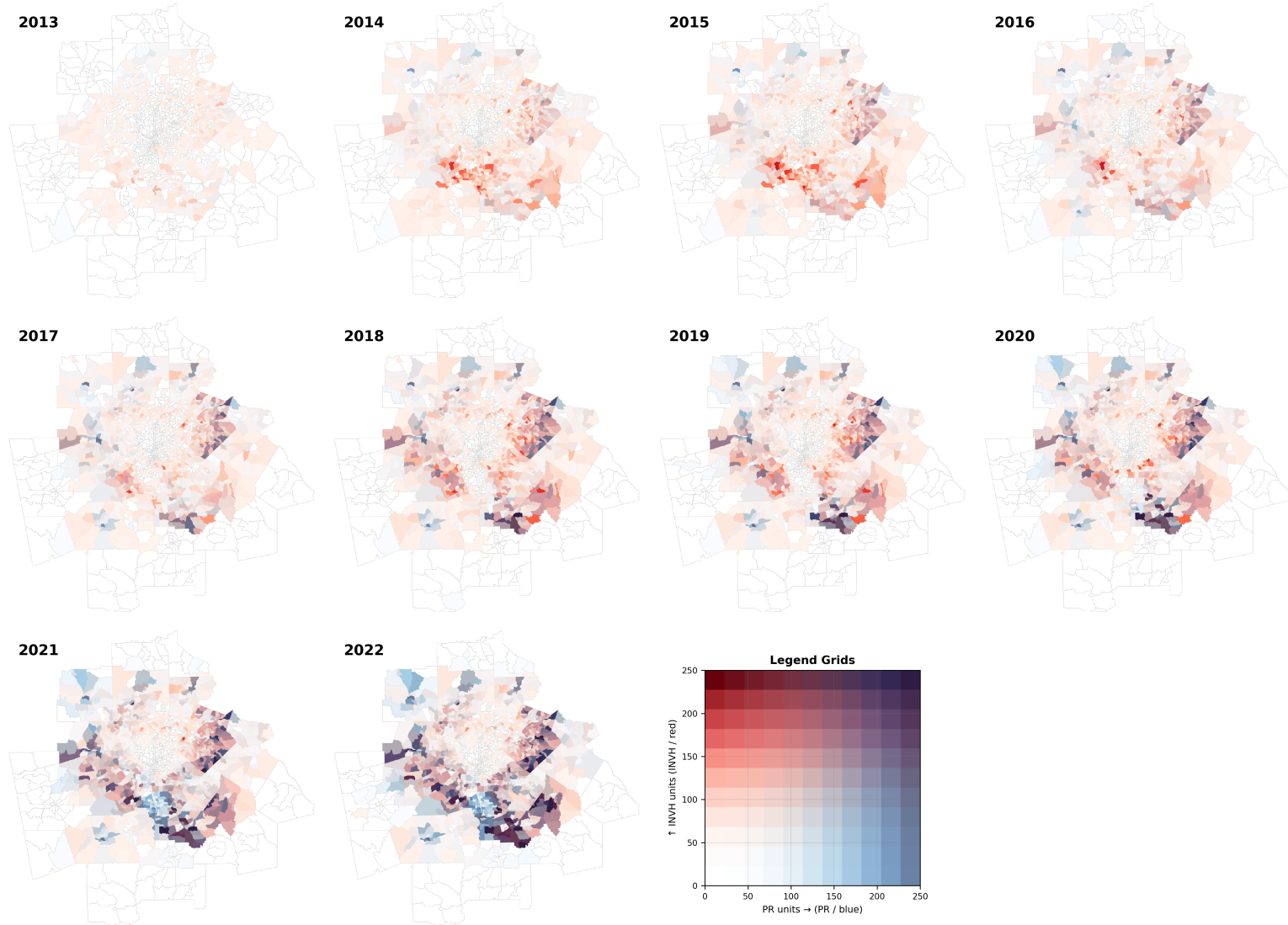
Local Density	[1,25)	[25, 50)	[50, 100)	[100,150)	[150, 200)	200+
2013	97.32	2.68	0.00	0.00	0.00	0.00
2014	70.62	21.81	7.13	0.44	0.00	0.00
2015	65.37	22.90	10.50	1.06	0.17	0.00
2016	64.50	22.76	11.00	1.56	0.18	0.00
2017	63.23	23.38	11.26	1.96	0.16	0.00
2018	50.48	24.09	18.43	4.53	1.84	0.63
2019	48.38	24.28	19.22	4.91	2.40	0.80
2020	45.20	25.09	20.37	5.89	2.19	1.25
2021	39.88	25.69	22.31	7.14	3.54	1.45
2022	34.25	24.16	24.33	9.21	4.59	3.47

Notes: We measure the local density for a home in a landlord's network using the number of total homes managed by the same landlord within 1 mile distance from the home of interest. We then categorize rental homes by how dense their local network is and display the share of each density range and its evolution across years.

Furthermore, institutional landlords differentiate in their choice of target locations to establish local densities. Figure 3 illustrates this point using examples of the two largest institutional landlords, Invitation Homes (Red) and Progress Residential (Blue), in the Atlanta metropolitan area. After decade-long expansion, we observe partial overlaps of two firms' portfolios alongside diversified clusters of areas dominated by one of the two firms. The spatial diversification suggests

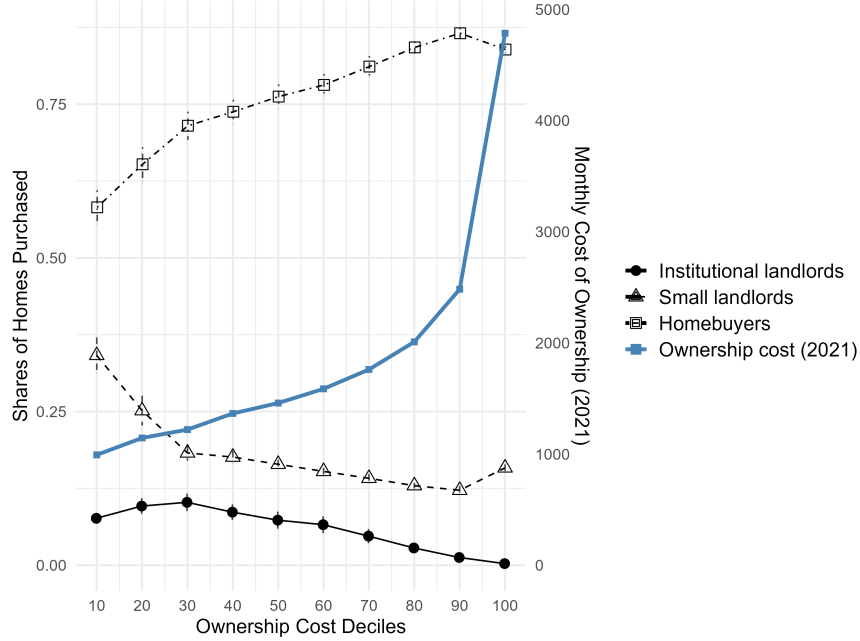
incentives to reduce direct competition in home acquisitions and to avoid direct rivalry with other institutional landlords in supplying similar rental products at the same or nearby locations.

Figure 3: Expansion Maps of Invitation Homes (Red) and Progress Residential (Blue), Atlanta MSA 2013-2022



Notes: We visualize the spatial expansion of the two largest institutional landlords in Atlanta metropolitan area, Invitation Homes (Red) and Progress Residential (Blue) from 2013-2022. Color gradients (from light to dark) represent the total number of units owned by the two firms at census tract level. Color temperature indicates the relative dominance in stock between two firms: warmer (red) tones indicate areas dominated by Invitation Homes, while cooler (blue) tones indicate areas dominated by Progress Residential.

Figure 4: Buyer Type Prevalence across Home Price Deciles, Atlanta MSA 2013-2021



Notes:

Fact 3. Institutional landlord rentals are primarily converted from homes in lower price ranges. Beyond spatial strategies, institutional landlords further concentrate their investment in lower-priced homes. Figure 4 shows the prevalence of three different types of buyers in home ownership transactions across homes' ownership cost deciles. Institutional landlords' acquisitions are concentrated in the lower-cost segments, with their presence peaking around the 20th–30th deciles and declining sharply thereafter. This pattern indicates several demand-driven incentives in landlords' investment strategies. Institutional landlords enter more in the segment of affordable houses, competing primarily with budget-constrained individual buyers, who tend to be more price-sensitive than households choosing among high-end housing options. Meanwhile, when converting into rentals, rental options of affordable houses can be more attractive to the group of budget-constrained households, compared to rentals offered to the higher-end of the market, where rentals are less of a substitute for purchase among wealthier households.

Fact 4: Institutional landlords' expansion is correlated to price, quantity, and participant composition changes in both the home ownership segment and the rental segment. We examine the correlational effects of institutional landlords' expansion on various housing market outcomes. Table 4 summarizes the regression results controlling for bedroom type $\times$ tract fixed effects and year fixed effects. In homeownership transactions, institutional landlords' presence is correlated to higher prices: a 100% increase of institutional buyers' share in a specific home type is associated with 30.4% increase in its sale price. Also, their purchases are correlated to

larger transaction volumes (supply of homeownership): one more unit purchased by institutional landlords is correlated to 0.744 more homes transacted on the market. Furthermore, the presence of institutional landlords correlates with compositional changes among non-institutional buyers. In home types entered more by institutional buyers, the reduction in individual homebuyers is greater than the decline observed among small landlord buyers.

In rental markets, one more unit bought by institutional landlords is correlated to 0.663 more units in total rental supply, regardless of landlord types. A 100% increase of institutional share in buyers is associated with 27.8 percentage point increase in institutional landlords' share in rental supply. The correlational effects on rental prices are smaller in scales. A 100% increase of institutional share in buyers of a specific home type is correlated to 3.23% decrease in its rental prices, if operated by small landlords. Given the same physical home type, institutional rentals on average are 3.95% higher in rental prices than its counterpart supplied by small landlords, indicating differences in rental service qualities. Suggested by this, in demand specification we model institutional landlords' rentals as differentiated products from their counterparts supplied by small landlords.

Table 4: Summary of regressions on housing market outcomes

Outcome Variables	Ownership Transactions			Rental Market		
	log(price)	transaction volume	homebuyer share in non-IL buyers	log(rent)	rental supply	Δ IL share in rental supply
Share bought by IL	0.304*** (0.0228)		-0.0364* (0.0158)	-0.0323** (0.0114)		0.278*** (0.0070)
Num. bought by IL		0.744*** (0.0496)			0.663*** (0.0240)	
Indicator: IL products				0.0395*** (0.0017)		
Bedroom $\times$ Tract FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Adj. R ²	0.626	0.823	0.358	0.699	0.344	0.478
Num. obs.	1,122,581	71,608	71,468	501,376	70,903	70,903
Level of obs.	unit	type	type	unit	type	type

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

4 Structural Model

This section presents the industry model depicting the evolution of a single family home market. Compared to previous models of housing markets, this paper develops a dynamic game model of rental supply, as investment is the key decision employed by institutional landlords to react to changing market conditions and competition landscapes. In the empirical dynamic games literature, this paper innovates by jointly modeling a static equilibrium of housing demand, supply, and price-forming conditional on institutional landlords' investment, and a dynamic equilibrium across periods that shapes investment paths and market evolution.¹⁰

4.1 Static Demand and Supply of Single-Family Homes

Market, product, and characteristics. We define the housing population in market mt as the mass of households entering or living in county m with no home ownership at the beginning of year t ¹¹. Products in the market are housing options h . Each inside option h aggregates single-family home units sharing physical characteristics, ownership types, and landlord types. We assume housing products are differentiated up to the home type level. Observable characteristics include: number of bedrooms, neighborhood walkable indicator, access to good schools, and PUMA location indicator. Ownership type distinguishes buying from renting. Landlord type is either small landlord or institutional landlord¹². The specification results in 792 single-family home types and 2,117 housing option types in Atlanta 2022.

Preferences. (Heterogeneous Coefficient Nested Logit model) Households make a discrete choice of housing across options from two nests $g \in \{0, 1\}$: for-rent options and for-purchase options. Household i 's utility from housing type h in market mt is given by:

$$u_{ihmt} = \alpha_{d(i)} p_{hmt} + X_{hmt} \beta_{d(i)} + \xi_{hmt} + \varsigma_{igmt} + (1 - \sigma) \epsilon_{ihmt} \quad (1)$$

where p_{hmt} denotes the monthly cost of housing option h . For homes for-sale, p_{hmt} includes mortgage interest, property insurance, and property tax payment conditional on home's transaction price¹³. For homes for-rental, p_{hmt} is the monthly rental payment measured using the median of rental listing prices. ξ_{hmt} denotes home type-level unobservable qualities. Finally, the household-

¹⁰The closest to this modeling approach are [Sweeting \(2013\)](#) in his study of radio stations and [Bodéré \(2023\)](#) in his study of preschools. Our framework differs in (i) allowing the supply structure of differentiated housing options to be shaped by firms' continuous choice of investment, instead of firms' discrete choices of formats (radio station) or quality (preschool); (ii) modeling multi-product firm strategies in the context with spatial differentiation.

¹¹We treat different counties in Atlanta as separate housing submarkets for two reasons. First, Georgia counties are geographically large, each encompassing a rich set of housing options. Second, Atlanta is recognized as a polycentric metropolitan area ([Goulbourne and Schuetz \(2021\)](#)). With multiple employment subcenters spread across the suburbs, local commuting patterns rarely converge exclusively on the Atlanta City and housing search happens more within the county of work or nearby.

¹²We assume homogeneous landlord services among rental products provided by all institutional landlords

¹³Appendix B.2 presents more details in ownership cost construction.

home type level shock is composed of group-level common shock ς_{igmt} and idiosyncratic shock ϵ_{ihmt} following type I extreme value distribution. We assume common nesting parameter across nests, $\sigma_g = \sigma, \forall g$, where $\sigma \in [0, 1)$. Households of different demographics $d(i)$ vary in coefficients for monthly costs and a subset of housing option characteristics - number of bedrooms, indicator for home ownership, and indicator for single-family homes. Specification is given by

$$\alpha_d = \alpha_0 + Z_d \pi^\alpha \quad (2)$$

$$\beta_d = \beta_0 + Z_d \pi^\beta \quad (3)$$

where demographics Z_d include household income, race and ethnicity, age, education, and number of children.

Given supply in each option S_h , household i 's probability of choosing type h is,

$$s_{h,d(i)} = \bar{s}_{h|g,d(i)} \bar{s}_{g,d(i)} = \frac{S_h \exp\left(\frac{\delta_{h,d(i)}}{1-\sigma}\right)}{D_{g,d(i)}^\sigma \left[\sum_g D_{g,d(i)}^{(1-\sigma)}\right]} \quad (4)$$

$$\text{where } D_{g,d(i)} = \sum_{h' \in g} S_{h'} \exp\left(\frac{\delta_{h',d(i)}}{1-\sigma}\right) \quad (5)$$

We denote the distribution of household demographics in the housing population as P_D and the size of housing population as M . The demand for type h aggregated from household choices is given by

$$D_h^{hhd} = M \cdot \int ds_{h,d(i)=z} dP_D(z) \quad (6)$$

Static supply. We model the supply of home ownership S^O and small landlord rentals S^{R-sl} as part of the static housing market equilibrium, conditional on the dynamic choices of investment by institutional landlords.

Supply of home ownership is composed of new constructions and units for-sale from the existing housing stock. Assume exogenous flow of new constructions, S_{hmt}^{O-new} . For supply from incumbent home owners, we model them as aggregations of individual potential seller's probability of selling given market conditions in mt and allow flexible patterns across incumbent owner occupiers and incumbent small landlords. The aggregated supply functions are

$$S_{hmt}^{O-oo} = g^{(1)}(p_{hmt}^O, \nu_{hmt}^{oo}; \theta^s) \cdot Q_{hmt}^{oo} \quad (7)$$

$$S_{hmt}^{O-sl} = g^{(2)}(p_{hmt}^O, p_{hmt}^{R-sl}, \nu_{hmt}^{sl}; \theta^s) \cdot Q_{hmt}^{sl} \quad (8)$$

For incumbent owner occupiers, we allow their selling probabilities to respond to home prices. This incorporates a home owner's incentive of capitalizing the property and relocate when the property value increases. For incumbent small landlords, we allow their selling probabilities to be jointly determined by $(p_{hmt}^O, p_{hmt}^{R-sl})$. This is micro-founded by an incumbent small landlord's optimal stopping problem where p_{hmt}^O and p_{hmt}^{R-sl} enter the exit value and continuation value respectively.

Similarly, we model the demand for home ownership from potential small landlord entrants as an aggregation of entry probabilities. Potential entrants are assumed to be tied to a given home type h , choosing between entering as a landlord or staying out. We assume the choice probabilities to be jointly determined by $(p_{hmt}^O, p_{hmt}^{R-sl})$. The aggregated demand is

$$D_{hmt}^{O-sl} = g^{(3)}(p_{hmt}^O, p_{hmt}^{R-sl}, \nu_{hmt}^{sl}; \theta^s) \cdot S_{hmt}^O \quad (9)$$

Net purchases by small landlords in year t update the stock of small landlord rentals in year $t + 1$,

$$Q_{hmt+1}^{R-sl} = Q_{hmt}^{sl} - S_{hmt}^{O-sl} + D_{hmt}^{O-sl} \quad (10)$$

where Q_{hmt}^{sl} represents the number of homes owned by small landlords when year t begins. In the rental sector, we assume zero vacancy rates for landlords of all types. Hence, $Q_{hmt+1}^{R-sl} = S_{hmt+1}^{R-sl}$.

Static market clearing. Conditional on institutional landlords' investment a_{hmt} and the resulting updated institutional landlord rentals, S_{hmt+1}^{R-il} , the static housing market equilibrium in mt is solved by a set of prices $\mathbf{P}_{mt} = \{(p_{hmt}^O, p_{hmt+1}^{R-sl}, p_{hmt+1}^{R-il})\}_{h \in \mathcal{H}}$ that clears demand and supply for each home type h across all ownership and landlord types.

$$S_{hmt}^{O-new} + S_{hmt}^{O-oo}(\mathbf{P}) + S_{hmt}^{O-sl}(\mathbf{P}) = D_{hmt}^{O,hhd}(\mathbf{P}) + D_{hmt}^{O-sl}(\mathbf{P}) + a_{hmt} \quad (11)$$

$$S_{hmt+1}^{R-sl}(\mathbf{P}) = D_{hmt+1}^{R-sl,hhd}(\mathbf{P}) \quad (12)$$

$$S_{hmt+1}^{R-il} = D_{hmt+1}^{R-il,hhd}(\mathbf{P}) \quad (13)$$

4.2 Institutional Landlords' Problem

On top of the static housing market described in Section 4.1, local housing stock evolves in its size and ownership structure, shaped by (i) new construction flows, (ii) ownership transactions between incumbent homeowners and incoming households and small landlords, and (iii) institutional landlords' investment over time. In this section, we present a dynamic equilibrium model of landlord investment, in which landlords are dynamic investors, adjusting their rental portfolios by choosing the number and types of homes to buy to maximize their aggregated values from the portfolio.

State space, actions, and timing. Landlords are indexed by j . We denote $(\mathcal{M}_{mt}, \xi_{mt})$ as the set of observed and unobserved market-level state variables common to all landlords. $\mathcal{M}_{mt}$

includes the size and demographic composition of the housing population, $(M, P_D)_{mt}$, the size and ownership structure of non-institutional housing stock $Q_{mt}^\lambda = \{Q_{hmt}^\lambda\}_{h \in \mathcal{H}, \lambda \in \{oo, sl, new\}}$, size of non-single family home stock Q_{0mt} , property-management wage w_{mt+1} , mortgage rates and property tax rates $(r_t^{\text{mtg}}, r_{mt}^{\text{tax}})$, and observable characteristics of a time-invariant grid of homes, $\{X_h\}_{h \in \mathcal{H}}$. ξ_{mt} represent the unobservable demand shocks $\{(\xi_{hmt}^O, \xi_{hmt+1}^{R-sl}, \xi_{hmt+1}^{R-il})\}_{h \in \mathcal{H}}$.

Apart from common states, landlord j faces landlord-specific state variables $(Q_{jmt}, Q_{-jmt}, \nu_{jmt})$. Q_{jmt} is an $H \times 1$ vector summarizing landlord j 's holdings of units in each home type $h \in \mathcal{H}$. Q_{-jmt} depicts the holdings of each competitor landlord $-j$. In each period, landlord j faces private action-specific shocks in investment cost, $\nu_{jmt}(a)$, capturing unobserved private costs associated with j 's investment process.

In year t , given existing stock Q_{jmt} and other state variables, landlord j chooses the number of homes to purchase or sell in each type h , denoted by $a_{jmt} = \{a_{jhmt}\}_{h \in \mathcal{H}}$. We model a_{jhmt} as a continuous choice variable taking values from $[-Q_{jhmt}, Q_{hmt}^{\text{total}} - Q_{jhmt}]$, where $Q_{hmt}^{\text{total}} = \sum_{\lambda \in \{oo, sl, new\}} Q_{hmt}^\lambda + \sum_j Q_{jhmt}$ represents the total housing stock in type h .¹⁴

We assume state realizations and actions follow the timing below:

1. Year t starts. Market-level states $(\mathcal{M}_{mt}, \xi_{mt})$ are realized. Landlord j begins with existing stock $\{Q_{jhmt}\}_{h \in \mathcal{H}}$ and observe all competitor landlords' stocks $\{Q_{-jhmt}\}_{h \in \mathcal{H}}$.
2. Private investment cost shocks are revealed to landlord j , $\nu_{jmt}(a)$. Landlord j chooses the number of units to purchase or sell across home types, $a_{jmt} = \{a_{jhmt}\}_{h \in \mathcal{H}}$, updating its rental stock to $Q_{jhmt+1} = Q_{jhmt} + a_{jhmt}$, $h \in \mathcal{H}$.
3. Housing market in t opens with landlord j 's choice included in the aggregate demand for ownership a_{mt} , which updates the supply of institutional landlord rentals S_{mt}^{R-il} together with other landlords' choices. Ownership and rental options are transacted at market-clearing prices $\mathbf{P}_{mt} = \{(p_{hmt}^O, p_{hmt+1}^{R-sl}, p_{hmt+1}^{R-il})\}_{h \in \mathcal{H}}$. Flow payoff for landlord j in year t Φ_{jmt} is realized. Non-institutional housing stock ownership structure evolves to

$$Q_{hmt+1}^\lambda = Q_{hmt}^\lambda - S_{hmt}^{O, \lambda} + D_{hmt}^{O, \lambda}, \quad \lambda \in \{oo, sl\}$$

4. Year t ends. Market-level exogenous state variables evolve to the next period following exogenous transition processes.

Flow Payoffs. In year t , landlord j faces investment costs composed of lump-sum payment for homes purchased in t and action-specific cost shocks,

$$c_{jmt}^I(a_{jmt}, \nu_{jmt}(a_{jmt})) = \sum_h a_{jhmt} p_{hmt}^O + \nu_{jmt}(a_{jmt}) \quad (14)$$

¹⁴Table 13 summarizes the frequency of a_{jhmt} values in sample.

where we make two assumptions on $\nu_{jmt}(a_{jmt})$ to facilitate estimation.

Assumption 1: $\nu_{jmt}(a_{jmt}) = \sum_h \nu_{jmt}(a_{hjmt})$

Assumption 2: $\nu_{jmt}(a_{hjmt}) = a_{hjmt} \cdot p_{hm0}^O \cdot \nu_{jhmt}$, $\nu_{jhmt} \sim^{\text{iid}} N(0, \sigma^\nu)$

Assumption 1 implies that action-specific cost shocks from the portfolio are additively separable across home types in the portfolio, with each term depending exclusively on within-type h 's investment level. Assumption 2 further implies that type-level investment shocks aggregate unit-level shocks, each proportional to the property's value in period 0. Intuitively we assume that property of higher values exhibits larger absolute volatility in its realized prices¹⁵.

By choosing investment a_{jmt} , landlord j adjusts its rental portfolio to Q_{jmt+1} and receives rental profits from year $t + 1$. We make a further assumption on landlord j 's supply of rental:

Assumption 3: $S_{jmt+1}^{R-il} = Q_{jmt+1}$

Assumption 3 implies a zero vacancy rate in landlord j 's rental stock: conditional on investment, landlord j are rental price takers, supplying every unit of its holdings to the rental market. Under Assumption 3, the model rules out the exercise of market power through pricing — that is, landlords are not allowed to withhold homes from the rental market to raise rents and increase profits.¹⁶ Landlord j 's rental profit is given by

$$\pi_{jmt+1} = \sum_{h \in \mathcal{H}} \left(p_{hmt+1}^{R-il} - \text{avc}_{jhmt+1} \right) Q_{jhmt+1} \quad (15)$$

where avc_{jhmt} is the average variable cost for j to manage a rental unit of type h in market $mt + 1$. To capture potential economies of scope and density in the provision of rental services, we model cost in two parts: the baseline cost and cost multipliers shifted by portfolio scope and density. We measure portfolio scope using the total number of homes operated by landlord j in county m ¹⁷. For each rental home h in the portfolio, we measure the local density using the total number of homes operated by landlord j in the same geographical subdivision (PUMA) as that of home h , $\mathcal{H}_{\text{puma}(h)} = \{h' \in \mathcal{H} : \text{puma}(h') = \text{puma}(h)\}$.¹⁸ To summarize, the average cost function is given by

¹⁵A standard assumption in asset pricing literature, see [Black and Scholes \(1973\)](#)

¹⁶According to public information disclosed by representative institutional landlords, their rental portfolios exhibit small vacancy rates. Quarterly reports from [Invitation Homes](#) and [American Homes 4 Rent](#) show that the unit-level occupancy rates range from 95-98% with seasonal fluctuations, exceeding the Georgia state average. This suggests a non-central role of mark-up to vacancy tradeoff in institutional landlords' strategies. By making assumption 3, we block the channel of pricing from all subsequent welfare calculations. [Barbieri and Dobbelt \(2025\)](#) study the margin of market power in pricing by institutional landlords and conclude limited presence of market power in this context.

¹⁷In Atlanta Metropolitan Areas, most administrative processes relevant to rental portfolio management - such as property taxation, eviction filings, code enforcement, and deed recording - are organized at the county level. We therefore use counties as the boundaries for landlord scope measurement. More regulatory details are available in the [O.C.G.A. § 44-2-1 \(2024\)](#) and [Georgia Landlord-Tenant Handbook](#), Georgia Department of Community Affairs.

¹⁸Figure XX displays the distribution of scale and density by landlord, county, and year.

$$avc_{jhmt+1} = \left(\gamma_0 + \gamma_1 w_{mt+1} + \gamma_2 p_{hmt}^{O, \text{lump sum}} \right) \cdot \left[\left(\frac{\tilde{Q}_{jmt+1}^{\text{total}}}{100} \right)^{\gamma_3} + \gamma_4 \left(\frac{\tilde{Q}_{hjmt+1}^{\text{local}}}{100} \right)^{\gamma_5} \right] \quad (16)$$

where

$$\tilde{Q}_{jmt+1}^{\text{total}} = \sum_{h' \in \mathcal{H}} Q_{jh'mt+1} \quad (17)$$

$$\tilde{Q}_{hjmt+1}^{\text{local}} = \sum_{h' \in \mathcal{H}_{\text{puma}(h)}} Q_{jh'mt+1} \quad (18)$$

where w_{mt+1} is the median wage of employees in the real estate leasing industry in county market $mt + 1$. $p_{hmt}^{O, \text{lump sum}}$ is the lump sum home ownership price for type h in mt . The baseline cost is modeled to capture labor inputs in the provision of rental services and the part of maintenance costs proportional to property's market value, including tax and insurance payments. Furthermore, the specification in Equation 16 assumes a constant rate of cost reduction across homes with different baseline costs, conditional on fixed portfolio scope and density.

To wrap up, landlord j 's payoff flow from year t is the sum of investment costs in t and discounted rental profits from $t + 1$.¹⁹

$$\begin{aligned} \Phi_{jhmt} (a_{jhmt}, a_{-jhmt}, Q_{jhmt}, Q_{-jhmt}, \mathcal{M}_{mt}, \xi_{mt}) = \\ - \sum_h p_h^O (a_{jhmt}, a_{-jhmt}, Q_{jhmt}, Q_{-jhmt}, \mathcal{M}_{mt}, \xi_{mt}) (1 + \nu_{jhmt+1}) \cdot a_{jhmt} \\ + 12 \times \beta \sum_h \left(p_h^{R-il} (a_{jhmt}, a_{-jhmt}, Q_{jhmt}, Q_{-jhmt}, \mathcal{M}_{\uparrow\downarrow}, \xi_{mt}) - avc_j (Q_{jhmt} + a_{jhmt}) \right) \cdot (Q_{jhmt} + a_{jhmt}) \end{aligned} \quad (19)$$

State transitions. We denote the full set of landlord j 's state variables as (s, ν) where s are the public state variables perceived by j and ν is j 's private action-specific shocks. The transition process $f(s_{jmt+1}, \nu_{jmt+1} \mid s_{jmt}, \nu_{jmt}, a_{jmt}, \sigma_{-j})$ is given by:

$$f(s_{jmt+1}, \nu_{jmt+1} \mid \cdot) = f(s_{jmt+1} \mid \cdot) f(\nu_{jmt+1}) \quad (\text{i.i.d. } \nu \text{ shocks}) \quad (20)$$

The transition of s_{jmt} is composed of multiple parts. The housing stock ownership structure transits following

¹⁹Note that landlord j 's rental revenue in $t + 1$ is determined with certainty conditional on landlord j 's investment choice in t , competitor landlords' choices in t , and housing market equilibrium solved in t . By including wage w_{t+1} in market-level state $\mathcal{M}_t$, landlord j 's rental profits from $t + 1$ is a function of a_{jt}, a_{-jt} and state variables in t only.

$$f(Q_{jmt+1}, Q_{-jm,t+1}, Q_{mt}^\Lambda | \cdot) = f(Q_{jmt+1} | a_{jmt}, Q_{jmt}) f(Q_{-jmt+1} | \cdot) f(Q_{mt}^\Lambda | \cdot), \text{ where} \\ f(Q_{jmt+1} | a_{jmt}, Q_{jmt}) = \mathbf{1}\{Q_{jmt+1} = Q_{jmt} + a_{jmt}\} \quad (21)$$

landlord j 's stock is updated deterministically conditional on the investment choice. (Q_{-j}, Q^{sl}, Q^{oo}) evolves following competitor's policy functions and the resulting equilibrium ownership transactions. We assume that landlord j expects the current corporate ownership of institutional landlords to persist and treat future merger events as unexpected exogenous shocks.²⁰ For new constructions, Q^{new} , we model the flow and its evolution as exogenous time series parameterized with θ^{trans} . Among the environment variables, all evolves exogenously following processes parameterized with θ^{trans} except the housing population variables, $(M, P_D)_{mt}$, which is jointly determined by the exogenous flow of net in-migration M_{mt+1}^0 and the endogenous flow of housing population retention from previous periods, with the latter shifted by the size and composition of renters solved in period t 's equilibrium.

$$M_{mt+1} = \eta \sum_h \left(D_{hmt+1}^{R-sl,hhd} + D_{hmt+1}^{R-il,hhd} \right) + M_{mt+1}^0 \quad (22)$$

where η is the (broad) retention rate of renters, defined as the probability for a renter household to join next year's housing demand in the same county²¹.

Optimality conditions. With revealed action-specific private shocks ν_j , landlord j 's optimal investment vector solves:

$$\max_{a_j} \mathbb{E}_{\nu_{-j}} [\Phi_j(a_j, \sigma_{-j}(s_{-j}, \nu_{-j}), s_j, \nu_j) + \beta \mathbb{E}[V_j^\sigma(s'_j) | a_j, \sigma_{-j}(s_{-j}, \nu_{-j}), s_j, \nu_j]] \quad (23)$$

where $a_j = (a_{1j}, a_{2j}, \dots, a_{Hj})$. Under strategy profile σ , we denote the ex-post value function as $V^\sigma(s, \nu)$ and ex-ante value function as $V^\sigma(s)$. In Equation 23, the outer expectation is taken over private shocks to competitors $-j$. The inner expectation is taken over shocks in state transitions to s' . The interior solution of a_{hj} satisfies

²⁰See similar assumptions in [Sweeting \(2013\)](#).

²¹In estimation, we estimate the empirical transition process of $(M, P_D)_{mt+1}$ as a whole, which is part of the equilibrium with presence and expansion of institutional landlords and changes little when landlord j makes marginal deviation in investment choices. See estimation notes in ???. In counterfactual study, we first calibrate the exogenous flows of $(M, P_D)_{mt+1}$ with parameter η , then construct counterfactual series of $(M, P_D)_{mt+1}$ for equilibrium banning institutional landlords from expansion.

$$0 = -p_{h0}^O \nu_{hj} + \mathbb{E}_{\nu_{-j}} \left[-p_h^O + \beta(p_h^R - \text{avc}_{jh}) - \mathbf{a}_j^\top \underbrace{\nabla_{a_{hj}} \mathbf{p}^O}_{\text{inflated cost of investment}} + (\beta(\mathbf{Q}_j + \mathbf{a}_j)^\top (\nabla_{a_{hj}} \mathbf{p}^R - \underbrace{\nabla_{a_{hj}} \mathbf{avc}_j}_{\text{cost benefits and spillovers}})) + \beta \frac{\partial EV_j^\sigma(s')}{\partial a_{hj}} \right] \quad (24)$$

Equality holds for all optimal choices of a_{hj} ranging between $-Q_{hj} < a_{hj} < Q_h^{\text{total}} - Q_{jh}$: landlord is not selling all its holdings in type h , and is not buying all units in the existing stock of type h .

Equilibrium concept. A Markov-Perfect Equilibrium of this model is,

1. a matrix of prices $\mathbf{P}_{mt} = \{(p_{hmt}^O, p_{hmt+1}^{R-sl}, p_{hmt+1}^{R-il})\}_{h \in \mathcal{H}}$
2. a matrix of housing stock by ownership $Q^\Lambda = \{Q_h^\lambda\}_{h \in \mathcal{H}, \lambda \in \{oo, sl, new\}}$ and institutional landlords' holdings, $\{Q_{hjmt}\}_{h \in \mathcal{H}, j \in \mathcal{J}}$
3. a policy function σ_j for each institutional landlord j in investment

such that,

1. given prices $\mathbf{P}_{mt}$ and housing supply $(S^O, S^{R-sl}, S^{R-il})_{hmt}$, households make their optimal housing choices, forming aggregated demand $(D^{O,hhd}, D^{R-sl,hhd}, D^{R-il,hhd})$.
2. given competitors' strategies σ_{-j} and realized states, each institutional landlord j optimally invest following policy function σ_j to maximize their value functions, forming aggregated demand D^{O-il} and updating rental supply to S^{R-il} ,
3. given prices $\mathbf{P}_{mt}$ small landlords decide optimally on entry/exit, aggregating to ownership demand D^{O-sl} , ownership supply S^{O-sl} and rental supply S^{R-sl} . Incumbent home owners decide optimally on selling/keeping their houses, aggregating to ownership supply S^{O-oo} .
4. prices $\mathbf{P}_{mt}$ clear the market:

$$S_{hmt}^{O-new} + S_{hmt}^{O-oo}(\mathbf{P}) + S_{hmt}^{O-sl}(\mathbf{P}) = D_{hmt}^{O,hhd}(\mathbf{P}) + D_{hmt}^{O-sl}(\mathbf{P}) + D_{hmt}^{O-il}$$

$$S_{hmt+1}^{R-sl}(\mathbf{P}) = D_{hmt+1}^{R-sl,hhd}(\mathbf{P})$$

$$S_{hmt+1}^{R-il} = D_{hmt+1}^{R-il,hhd}(\mathbf{P})$$

5 Estimation

We conduct estimation of the structural model in two general steps. First, we use market-level data with micro moments from all county-year's ownership and rental segments to estimate demand

parameters in the heterogeneous coefficient nested logit model (Berry, 1994; Berry et al., 1995; Nevo, 2000, 2001; Petrin, 2002; Grigolon and Verboven, 2014; Conlon and Gortmaker, 2025). Second, based on demand estimates, static supply estimates, and transition process estimates, we adapt the BBL two-step framework to the continuous choice context and obtain MLE estimator for cost parameters (Hotz et al., 1994; Ackerberg et al., 2007; Bajari et al., 2007).

5.1 Estimating Housing Demand Parameters

Identification. There exist three sources of endogeneity in the housing demand system. First, endogenous prices $\mathbf{P}_{mt}$, rooted from their correlation to unobserved quality ξ_{mt} . In sorting equilibrium, holding all else fixed, market-clearing prices tend to be lower for housing products with unobservably lower qualities. Second, endogenous supply of housing in both ownership segment S^{O-oo}, S^{O-sl} and rental segment S^{R-sl} . For supply determined by local small landlords and incumbent owner-occupiers, their probabilities of supply are functions of endogenous prices. For supply determined by institutional landlords, the unobserved qualities ξ_{mt} are part of their state variables, shifting their investment choices and in turn, their rental supply. Third, the endogenous within-nest shares of housing types $s_{h|g}$. Within the same nest and all else fixed, housing types with higher unobserved qualities will have higher within-nest shares.

To address the price endogeneity, we use an identification strategy that leverages the cross-market variation in supply structure across housing types, following Berry et al. (1995) and Bayer et al. (2007). The relevance condition holds, because holding all else fixed, a house will have a higher price in the sorting equilibrium if the supply of its close substitutes is more scarce. Given endogenous supply in our context, we further refine the strategy by replacing real supply terms with fitted values of supply, predicted by exogenous variables only. To conclude, the conditions used to identify linear demand parameters are:

$$\forall h : g_h(\tilde{\mathbf{P}}, X^{\text{exog}}, \tilde{S}, \xi = 0, P_D; \theta^D) = s_h \quad (25)$$

$$\mathbb{E}[\xi \tilde{\mathbf{P}}] = 0 \quad (26)$$

$$\mathbb{E}[\xi X^{\text{exog}}] = 0 \quad (27)$$

where g_h is the expectation of households' aggregated share of choosing type h under candidate demand parameter θ^D , s_h is the observed share of type h aggregated from households' choices in data. Equation 25 specifies a model-generated instrument $\tilde{\mathbf{P}}$ that solves the real-world market shares with two key differences: $\xi = 0$ and exogenous fitted value of supply, $\tilde{S}$. By construction, $\tilde{\mathbf{P}}$ satisfies the exclusion restriction and is a valid instrument for price.

To identify the nesting parameter, we use home type-level housing stock as an exogenous instrument to shift within-nest shares. Identification leverages the cross-market variation attributed

to new constructions and changes in outside options. We denote house type h as $(\bar{h}, k(h))$, where $\bar{h}$ represents the physical type h belongs to, and $k(h) \in \{O, R - sl, R - il\}$ represents the owner group. First, $\hat{Q}_{hmt}^{IV}$ is given by $\hat{Q}_{hmt}^{IV} = Q_{hmt}^{total} \cdot \gamma_m^{k(h)}$, where $\gamma_m^{k(h)}$ is the mean rate of supply by owner group $k(h)$ in county m over all types and periods.

$$\hat{s}_{h|g,mt}^{IV} = \frac{\hat{Q}_{hmt}^{IV}}{\sum_{h' \in g} \hat{Q}_{h'mt}^{IV}} \quad (28)$$

The moment is given by

$$\mathbb{E}[\xi \hat{s}_{h|g}^{IV}] = 0 \quad (29)$$

Apart from endogeneity issues, we further recover households' heterogeneity in tastes by demographics using correlation between households' choice characteristics and households' demographics²².

Implementation. We construct the model-generated instrument for prices $\tilde{\mathbf{P}}_{mt}$ as follows:

1. Prepare a grid of candidate demand parameters, $\tilde{\Theta}$. In market mt , prepare exogenously predicted supply structure $\{\tilde{S}_h\}_{h \in \mathcal{H}}$, housing option characteristics $\{X_h\}_{h \in \mathcal{H}}$, and size and distribution of housing population, (P_D)
2. Set $\xi = 0$. For each $\theta^D \in \tilde{\Theta}$, solve for prices $\tilde{\mathbf{P}}_{mt}(\theta^D)$ such that $g_h(\tilde{\mathbf{P}}_{mt}, X_{mt}, \tilde{S}_{mt}, \xi_{mt} = 0, P_D; \theta^D) = s_{hmt}$ for all $h \in \mathcal{H}$, where s_{hmt} is the observed share of housing type h formed by households choices only in market mt .

We then use the nested fixed point algorithm to search for the GMM estimator of demand parameters, $\theta^D = (\alpha, \beta, \pi^\alpha, \pi^\beta, \sigma)$. The outer loop searches for nonlinear parameters, $(\pi^\alpha, \pi^\beta, \sigma)$. For each guess of nonlinear parameters, we calculate the demographic-specific parts of the utility and invert market shares to recover the mean utility of each housing type. Given mean utility, in the inner loop we obtain linear IV estimates of (α, β) (conditional on the guess of nonlinear parameters). Instruments used include exogenous housing type characteristics, X^{exog} and price instruments, $\{\tilde{\mathbf{P}}_{mt}(\theta^D)\}_{\theta^D \in \tilde{\Theta}}$. We use an initial guess of linear parameters and locate the closest candidate on grid, $\theta_0^{\text{grid}-D}$ and the corresponding instrument $\tilde{\mathbf{P}}_{mt}(\theta_0^{\text{grid}-D})$. Using this, we obtain the IV estimates θ_1^{IV-D} and use it to update the closest candidate on grid, $\theta_1^{\text{grid}-D}$, and iterate... until the difference between iterations closes to below tolerance criteria²³. Finally, we calculate the correlation using simulated data, the IV moment for nesting parameter, and the updated GMM objective function value.

Results. We use data from 20 housing markets in the Atlanta metropolitan area 2011-2021 for demand estimation. Table 5 presents the GMM estimates for (α, β, σ) and Table 6 for (π^α, π^β) .

In progress: visualization of demand elasticity by demographics and interpretation writing.

²²Table 14 summarizes the correlation between selected demographics and housing characteristics.

²³Table ?? presents first-stage performance of the model-generated instruments.

Table 5: Parameter Estimates for Single-Family Home Demand

	Estimates	SE
Monthly cost (\$1,000)	-11.070	0.371
Three-bedroom	2.150	0.089
Four-bedroom	4.825	0.200
Five and more-bedroom	8.485	0.367
Walkable neighborhood	-0.356	0.060
One high-quality school	1.127	0.060
Two high-quality schools	1.668	0.081
Three high-quality schools	2.272	0.107
Indicator: branded landlord	1.483	0.056
Indicator: ownership	-8.551	0.158
FEs: geographical division (PUMA)	X	
FEs: county market SFH indicator	X	
σ	0.485	0.650

Notes: Figures represent taste parameters of the default demographic group (mean log(income), no child, younger than 25 yrs old, no bachelor degree, Race: Others).

5.2 Estimating Other Static Supply and Demand Parameters

We parameterize the static ownership supply functions and small landlords' ownership demand function in Section 4.1 using probit links,

$$S_{hmt}^{O-oo} = \Phi(\log(p_{hmt}^{O,lump\ sum}), FE_h, \nu_{hmt}^{O-oo}; \theta_1^S) \cdot Q_{hmt}^{oo} \quad (30)$$

$$S_{hmt}^{O-sl} = \Phi(\log(p_{hmt}^{O,lump\ sum}/p_{hmt+1}^{R-sl}), FE_h, \nu_{hmt}^{O-sl,1}; \theta_2^S) \cdot Q_{hmt}^{sl} \quad (31)$$

$$D_{hmt}^{O-sl} = \Phi(\log(p_{hmt}^{O,lump\ sum}/p_{hmt+1}^{R-sl}), FE_h, \nu_{hmt}^{O-sl,2}; \theta_3^S) \cdot S_{hmt}^O \quad (32)$$

where $\Phi(\cdot)$ is the cumulative distribution function (henceforth cdf) of the standard normal distribution. Parameters of interest are $(\theta_1^S, \theta_2^S, \theta_3^S)$. Prices and rents are endogenous given their correlation to unobservables in the supply system, $\nu^{O-oo}, \nu^{O-sl,1}, \nu^{O-sl,2}$. We leverage instruments that exogenously shift the prices and rents from the demand side, and specifically the variation in demographic distribution in housing population across markets.²⁴ Table 12 summarizes the first-stage regression results of these instruments. The F statistic is 58.85 for estimating Equation 30 and is 13.04 for estimating Equation 31 and 32. Table 7 presents the estimates of supply parameters with home

²⁴As defined earlier, housing population refers to the group of households who enter a county's market in a given year with no home ownership. Therefore, their demographic attributes shift prices and rents only from the demand side. For Equation 32, unobservables exist in potential small landlord entrants' demand functions. Demographic instruments are valid too, under the assumption that housing population demographics are exogenous to potential small investors' attributes.

Table 6: Non-linear Parameter Estimates for Single-Family Home Demand

	Monthly cost (\$1,000)	Three- bedroom	Four- bedroom	Five- bedroom	Indicator: ownership	Indicator: SFH
log(income)	0.422 (0.281)	0.063 (0.082)	0.143 (0.161)	0.511 (0.737)		-0.454 (0.495)
Income group: \$50k-\$100k					0.932 (0.432)	
Income group: \$100k-\$150k					2.014 (0.484)	
Income group: \$150k-\$200k					2.678 (0.490)	
Income group: \$200k+					3.334 (0.629)	
Number of child: 1-2	0.843 (0.749)	0.116 (0.284)	0.158 (0.331)	-0.107 (0.410)	0.996 (0.437)	1.123 (2.238)
Number of child: > 2	1.198 (1.125)	0.046 (0.325)	0.533 (0.587)	0.519 (0.627)	1.223 (0.909)	1.942 (2.324)
Age: 25-35	-0.233 (1.119)				-0.029 (1.213)	-0.592 (1.408)
Age: 35-45	-0.111 (1.223)				-0.252 (1.236)	0.243 (1.789)
Age: 45-55	-0.013 (1.120)				0.283 (1.256)	0.616 (2.078)
Age: 55+	-0.221 (1.049)				0.588 (1.200)	0.729 (1.688)
Bachelor's Degree	1.470 (2.355)				2.520 (2.181)	
Race: White, Non Hispanic	1.199 (1.831)				1.512 (2.603)	
Race: Black	-2.037 (6.664)				-2.812 (7.849)	
Race: White, Hispanic	-0.367 (4.724)				-0.120 (6.846)	
Race: Asian	1.040 (1.873)				2.037 (2.685)	

type fixed effects controlled. For an incumbent owner occupier, the probability of selling increases by 0.125 given 1% increase in total sale prices. Holding rental prices as fixed, the probability of selling from incumbent small landlords increase by 0.337 given 1% increase in total sale prices. The same price change, to the potential small landlord entrants, will decreases their probability of buying by 2.251. Based on the results, we calculate ownership supply elasticities across home types and plot the distribution in Figure 5.²⁵ Overall, the ownership supply from existing homes exhibits low elasticity, with notable heterogeneity across sellers. Among incumbent owner-occupiers—the majority of potential sellers—the median supply elasticity is 0.253. Incumbent small landlords display relatively higher elasticity, with a median of 0.659, with greater variance across home types

²⁵For small landlords we hold rental prices as fixed and only make marginal changes to the lump sum ownership prices.

Table 7: Other Static Supply and Demand Parameters

	Coefficients	Estimates	SE
<i>Supply of ownership from OO, S^{O-oo}</i>			
$\log(p_{hmt}^{O, \text{lump sum}})$	θ_1^S	0.125	0.025
FEs: home type		X	
<i>Supply of ownership from SL, S^{O-sl}</i>			
$\log(p_{hmt}^{O, \text{lump sum}}/p_{hmt+1}^{R-sl})$	θ_2^S	0.337	0.057
FEs: home type		X	
<i>Demand of ownership from SL, D^{O-sl}</i>			
$\log(p_{hmt}^{O, \text{lump sum}}/p_{hmt+1}^{R-sl})$	θ_3^S	-2.251	0.156
FEs: home type		X	

Notes: We measure the total ownership price of houses, $p^{O, \text{lump sum}}$ in \$1,000, not including property taxes, mortgage interests, or insurances. We measure the rental prices of homes supplied by small landlords, p^{R-sl} using the monthly rents in \$.

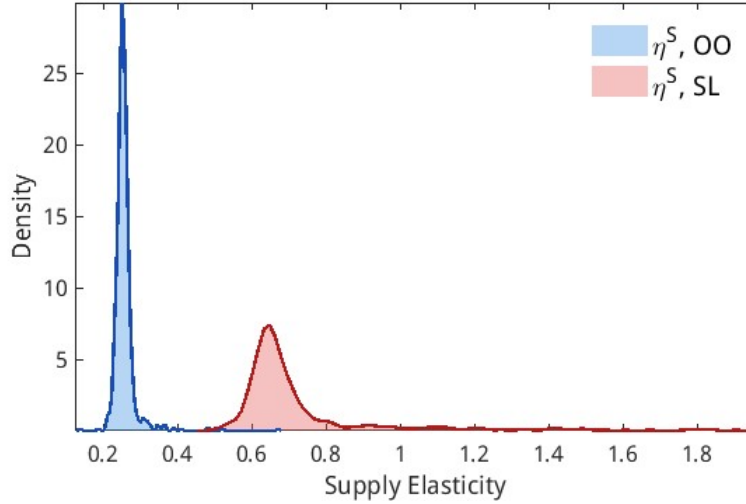


Figure 5: Distribution of ownership Supply Elasticity across Home Types

compared to owner-occupier sellers.

5.3 Estimating Institutional Landlord Cost Parameters

We exploit landlords' interior choices of investment a_{hjmt} under varying states to identify the dynamic parameters $\theta^C = (\gamma, \sigma_\nu)$ in their cost functions. Recall Equation 24:

$$0 = -p_{h0}^O \nu_{hj} + \mathbb{E}_{\nu_{-j}} \left[-p_h^O + \beta(p_h^R - \text{avc}_{jh}) - \mathbf{a}_j^\top \underbrace{\nabla_{a_{hj}} \mathbf{p}^O}_{\text{inflated cost of investment}} + (\beta(\mathbf{Q}_j + \mathbf{a}_j)^\top (\nabla_{a_{hj}} \mathbf{p}^R - \underbrace{\nabla_{a_{hj}} \mathbf{avc}_j}_{\text{cost benefits and spillovers}})) + \beta \frac{\partial EV_j^\sigma(s')}{\partial a_{hj}} \right]$$

The challenging part in landlord's optimality condition is $\mathbb{E}_{\nu_{-j}} \left[\frac{\partial EV_j^\sigma(s')}{\partial a_{hj}} \right]$, which summarizes the expected marginal effect of a_{hj} chosen in one period on the expected continuation value evaluated in the next period. In the multi-agent dynamic game, a full solution of value function is not feasible. Therefore, we follow the BBL two-stage framework to first propose a method of forward-simulation, then apply the method to calculate the $\mathbb{E}_{\nu_{-j}} \left[\frac{\partial EV_j^\sigma(s')}{\partial a_{hj}} \right]$ at states of interest. Finally, we assemble the results and develop a maximum likelihood estimator using Equation 24.

First Stage: Neural Network Estimation of Policy Functions. With sufficiently large number of observations, policy functions would be estimated nonparametrically. However, this is not feasible given the dimension and size of the state space with continuous state variables, compared to which the sample size is relatively small. To overcome the challenge, we estimate policy functions using neural network in combination with parametric restrictions to facilitate a balance between flexible functional form and reasonable scale of predictions made at both visited and not-visited states.

The policy function, $\sigma_j(\cdot)$ is a mapping from firm j 's state variables to the (dis)investment choice on H differentiated home types. We re-write the choice for each dimension as the product of policy ratio (r_{hgmt}^a) and stock of homes owned by non-institutional landlords. The policy ratio is allowed to go negative.

$$a_{hjmt} = r_{hgmt}^a \cdot (Q_{hmt}^{oo} + Q_{hmt}^{sl} + Q_{hmt}^{new}) \quad (33)$$

Then, we train a neural network model to predict r_{hgmt}^a as a function of a rich set of inputs composed of state variables and functions of state variables. We denote the data generating process of policy ratio as

$$r_{hgmt}^a = g(x_{hjmt}, \nu_{hjmt}) \quad (34)$$

, where ν_{hjmt} enters $g(\cdot)$ linearly with $g_\nu < 0$, as guaranteed by landlord j 's optimality condition in Equation 24 under Assumptions 1 and 2. We denote the trained neural network model as $\hat{g}^{mnt}(x_{hjmt}; \hat{\theta}^g)$, which predicts the ex-ante expectation of policy ratio that would be chosen by firm j given some realized value of x_{hjmt} , before the realization of cost shocks ν_{hjmt} . To recover the empirical distribution of residual investment driven by the cost cost shocks, we calculate the

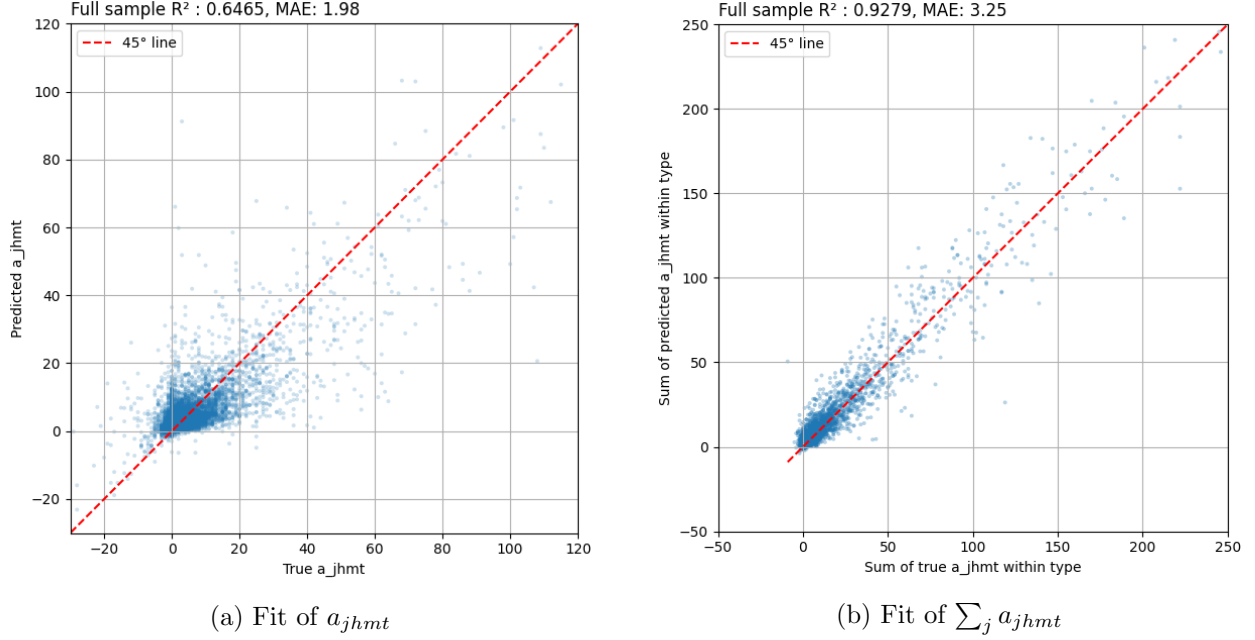


Figure 6: First-stage Fitted Policy Functions $\hat{\sigma}_j$, Neural Network with 2 Layers of 128/64 Units

residuals

$$\hat{\epsilon}_{hjmt}^a = r_{hjmt}^a - \hat{g}^{nnt}(x_{hjmt}; \hat{\theta}^g) \quad (35)$$

and use the sample variance of $\hat{\epsilon}_{hjmt}^a$ as the sample analog to the variance of the distribution of shock-driven residual investment. The neural network estimates, together with the estimated variance of residuals, will be used in the forward simulation method of approximating value functions to predict firms' investment choices under different realizations of states and shocks. In simulation, investment shocks $\nu_{hjmt} = \nu^*$ and residual investment $\epsilon_{hjmt}^a = \epsilon^*$ are drawn in a joint way, such that

$$F_{\epsilon^a}(\epsilon^*) = 1 - F_{\nu}(\nu^*) \quad (36)$$

where F_{ϵ^a} is the cdf of residual investment and F_{ν} is the cdf of investment shock basis.

Figure 6 displays the full sample fit of neural network model with 2 layers of 128/64 units. Panel (a) shows the model fit of firm-home type level investment, comparing $\mathbb{E}_{\nu}[\hat{\sigma}_{jh}(s_{jmt}, \nu)]$ against $a_{jh} = \sigma_{jh}(s_{jmt}, \nu_{jhmt})$ under some realized value of private shock ν_{jhmt} . Distance from each point to the 45-degree line is the sum of (i) potential model specification error in $\hat{\sigma}_j$ and (ii) private-shock driven residual investment aggregated out when taking expectation. Panel (b) further shows that, when aggregating investment choices over firms in the same market, year, and home type, distances between predicted expectations and real data are notably shortened, with R^2 increases from 0.647 to 0.928. The diminishing distance is mainly driven by part (ii): idiosyncratic firm-type investment shocks aggregate out in sample sums with 5-10 firms, leaving remaining distance primarily attributed to estimation or model specification error in $\hat{\sigma}_j$. Suggested by the good fit in

Panel (b), neural network estimation of $\hat{\sigma}_j$ provides reliable first-stage fit of firms' policies.

First Stage: Forward Simulation and Value Function Approximation. Using estimated policy functions $\hat{\sigma}_j$ and state transition processes $\hat{\theta}^{trans}$, we forward-simulate equilibrium paths starting from a state of interest, which later, will be used to approximate the value function values. Unlike many forward-simulation practices that involve candidate dynamic parameters, in this paper the simulated paths are not dependent on the value of dynamic parameters, instead only use estimates from demand, static supply, state transitions, and first-stage fitted policy functions. This significantly reduces the computation burden, allowing second stage estimation to search and evaluate candidate parameters using a fixed set of inputs from forward simulation.

We follow the steps below to approximate $\beta \mathbb{E}_{\nu-j} \left[\frac{\partial EV_j^\sigma(s')}{\partial a_{hj}} \right]$.

1. Assume the state of interest is s_{0j} and the optimality condition is derived from the choice of investment by j in type h^* .
2. Starting from $t = 0$, we simulate $N^{\text{sim}} = 500$ paths of length $T^{\text{sim}} = 20$, in which exogenous state variables evolve following transition processes with realized shocks. In $t = 0$, landlord j chooses the investment vector a_{jmt} observed in data. Other landlords $-j$ draw their investment levels following $\hat{\sigma}_{-j}(s_{0,-j})$ and the cdf of realized private shocks on each path. We solve the static housing equilibrium and update the housing ownership structure to next period.
3. For $t = 1, \dots, T^{\text{sim}}$, all landlords invest follow $\hat{\sigma}$ and the cdf of the realized private shocks. For each period on the simulation path, we save the equilibrium prices, landlord j 's investment choices, private cost shocks, and updated rental stock.
4. Repeat steps 2 and 3 with one change in $t = 0$. In $2'$, landlord j chooses the investment vector with Δa units of deviation from the choice in data in home type h , $a'_{hjmt} = a_{hjmt} + \Delta a$ only for $h = h^*$. We specify $\Delta a = 1$. Simulate the subsequent equilibrium path given updated choice of investment in h^* by landlord j .
5. Given the two sets of simulation results and candidate parameter θ^C , calculate

$$\beta \bar{V}(s' | a_j; \theta^C) = \frac{1}{N^{\text{sim}}} \sum_{i=1}^{N^{\text{sim}}} \sum_{t=1}^{T^{\text{sim}}} \beta^t \Phi_j(a_{jt}, Q_{jt+1}, p_t^O, p_{t+1}^{R-il}, w_{t+1}, \nu_{jt}; \theta^C) + \beta^{T^{\text{sim}}} V^{\text{last}}(Q_{jT+1}, p_T^O, p_{T+1}^{R-il}, w_{T+1}; \theta^C) \quad (37)$$

$$\beta \bar{V}(s' | a_j + \Delta a_{h^*}; \theta^C) = \frac{1}{N^{\text{sim}}} \sum_{i=1}^{N^{\text{sim}}} \sum_{t=1}^{T^{\text{sim}}} \beta^t \Phi_j(a'_{jt}, Q'_{jt+1}, p_t^{O'}, p_{t+1}^{R-il'}, w_{t+1}, \nu_{jt}; \theta^C) + \beta^{T^{\text{sim}}} V^{\text{last}}(Q'_{jT+1}, p_T^{O'}, p_{T+1}^{R-il'}, w_{T+1}; \theta^C) \quad (38)$$

where the V^{last} is the continuation value evaluated at T^{sim} as the present discounted value of all future rental profits, assuming no more adjustment to rental stock by any of the institutional landlords starting $T^{\text{sim}} + 1$. We approximate the discounted expectation of value function gradient using:

$$\beta \frac{\partial \bar{V}_j(s'; \theta^C)}{\partial a_{h^*j}} = \frac{\beta \bar{V}(s' | a_j + \Delta a_{h^*}; \theta^C) - \beta \bar{V}(s' | a_j; \theta^C)}{\Delta a_{h^*}} \quad (39)$$

Second Stage: Optimality Condition and MLE Estimator. For each (s_{jmt}, a_{jhmt}) in data satisfying interior solution condition, $-Q_{jhmt} < a_{jhmt} < Q_{hmt}^{\text{total}} - Q_{jhmt}$, given candidate parameter $\theta^C = (\gamma, \sigma_\nu)$, we can recover the private investment cost shock using sample analog of Equation 24:

$$\hat{\nu}_{hj} = \frac{1}{p_{h0}^O} \cdot \left[\frac{1}{N^{\text{sim}}} \sum_{i=1}^{N^{\text{sim}}} -p_{h,i}^O + \beta(p_{h,i}^R - \text{avc}_{jh,i}) - \mathbf{a}_j^\top \nabla_{a_{hj}} \mathbf{p}_i^O + \beta(\mathbf{Q}_j + \mathbf{a}_j)^\top (\nabla_{a_{hj}} \mathbf{p}_i^R - \nabla_{a_{hj}} \mathbf{avc}_{j,i}) \right] + \beta \frac{\partial \bar{V}_j(s')}{\partial a_{hj}} \quad (40)$$

For notation convenience, we denote the recovered $\hat{\nu}_{hj}$ as a function of Data_{jh} and candidate parameters (γ, σ_ν) . By Assumption 2, ν_{hj} follows a normal distribution with unknown variance, $N(0, \sigma_\nu^2)$. The negative log-likelihood function is given by

$$\text{NLL}(\gamma, \sigma_\nu) = \frac{N}{2} \log(2\pi\sigma_\nu^2) + \frac{1}{2\sigma_\nu^2} \sum_{i=1}^N [\hat{\nu}_{hj}(\text{Data}_i; \gamma, \sigma_\nu)]^2 \quad (41)$$

Results. Table 8 presents results from dynamic parameter estimation. Estimates show that in the baseline cost, labor costs weigh more than the property-value related costs. Estimates quantify the economies of scope and density. $\hat{\gamma}_3 < 0$ and $\hat{\gamma}_4 \cdot \hat{\gamma}_5 < 0$, indicates that average variable costs of maintenance decreases as portfolio size and density grows. Given $\hat{\gamma}_3 < 0$ and $\hat{\gamma}_5 < 0$, the marginal cost reduction to scope and density are both diminishing. Hence, marginal cost reduction is smaller for firms expanding from a larger and denser site.

6 Counterfactual Analysis

We present counterfactual analysis in two parts. First, we quantify and decompose the distributional effects of institutional landlords on social welfare. Second, we show the role of endogenous cost efficiency played in driving the expansion when institutional landlords are forward-looking.

6.1 Distributional Effects on Social Welfare

To estimate the effects of institutional landlord's expansion on welfare distribution, we conduct a counterfactual analysis where institutional landlords are prohibited from expansion from 2013 and

Table 8: Dynamic Parameters

	Coefficients	Estimates
<i>Baseline Costs</i>		
Intercept	γ_0	779.414
Wage	γ_1	0.142
Property value (in \$1,000)	γ_2	0.011
<i>Cost Multipliers</i>		
Economies of scope, nonlinear	γ_3	-0.286
Economies of density, linear	γ_4	0.439
Economies of density, nonlinear	γ_5	-0.214
<i>Private Cost Shocks</i>		
Std. Dev.	σ^ν	0.203
<i>Statistics</i>		
Negative Log Likelihood		0.393
Observations		9,929

Notes: estimation is conducted setting $\beta = 0.92$ and $\alpha^{\text{retention}} = 0.8$. Standard Errors in progress.

on. We simulate the housing market equilibrium from 2013 (Year 0) to 2021 (Year 8), excluding institutional landlords from home ownership transactions. We recompute market-clearing prices for ownership options and rental options, quantities transacted, and the equilibrium path of housing ownership structure evolution.

Households. With nested logit demand specification, household i 's compensation variation is given by

$$CV_{it} = \frac{12}{-\alpha_{d(i)}} \left(\sigma \ln \sum_g \left(U_{igt}^{\text{With IL}} \right)^{\frac{1-\sigma}{\sigma}} - \sigma \ln \sum_g \left(U_{igt}^{\text{No IL}} \right)^{\frac{1-\sigma}{\sigma}} \right) \quad (42)$$

where CV_{it} is measured in year t 's dollar. The inclusive value from nest $g \in \{0, 1\}$ is given by:

$$U_{igt} = \sum_{h \in g} \exp \left(\frac{\bar{u}_{d(i)ht}}{1 - \sigma} \right) \quad (43)$$

where $\bar{u}_{d(i)ht} = \alpha_{d(i)} p_{ht} + X_{ht} \beta_{d(i)} + \xi_{ht}$, denotes the deterministic part of utility that household of demographic $d(i)$ receives from home type h in t .

We separate welfare calculation for households conditional on their choice of renting versus buying in the real world with institutional landlords' expansion.

Table 9 displays consumer welfare changes by households' choice of renting versus buying.

Institutional Landlords. Using cost function estimates, we recover the firm-property level maintenance costs and their evolution across years. Figure 7 (a) plots the trend of average variable

Table 9: Welfare Changes by Year and Households' Tenure Choice

	2014	2015	2016	2017	2018	2019	2020	2021
Share of Renters (%)								
(with IL expansion)	92.28	91.87	91.54	91.86	92.29	92.01	91.76	92.60
(no IL expansion)	91.40	91.23	91.01	91.20	91.53	91.19	90.99	91.01
(Changes)	0.88	0.64	0.53	0.66	0.76	0.82	0.77	1.58
Number of Households								
(with IL expansion)	539,065	551,753	566,209	571,211	585,472	618,384	623,262	642,496
(no IL expansion)	533,808	544,102	557,783	562,688	576,255	608,061	611,687	630,242
(Changes)	5,257	7,651	8,425	8,523	9,216	10,323	11,576	12,255
ΔCS , Renters								
(per household)	19.50	34.14	41.77	52.33	56.04	60.62	66.39	64.94
ΔCS , Buyers								
(per household)	-155.14	-51.38	-10.54	1.17	-12.36	19.88	16.04	-85.53

costs, standardized as a fraction of the cost of running one isolated rental unit with no scope or density economies. Along with portfolio expansion, institutional rentals' mean average variable cost decreases from 0.33 to 0.20 with a shrinking standard deviation from 0.11 to 0.05. Figure 7 (c) plots the trend of rental profits per property month. As institutional landlords expand, average rental profits per unit month increase from \$-235.72 to \$300.95 with a shrinking standard deviation from 536.62 to 329.40. In both plots we provide the paths of two firms, Invitation Homes (red dashed) and Amherst (blue dot dashed), to illustrate firm heterogeneity. Given diminishing returns to scope and density indicated by $\hat{\gamma}_3, \hat{\gamma}_5 < 0$, marginal cost reduction is smaller for firms expanding from a larger and denser base. Invitation Homes begins with a larger portfolio size and keeps lower-than-industry average costs throughout the years with relatively less reduction in costs. Amherst, in contrast, begin with a smaller portfolio size and high costs, reduces its costs at a relatively fast pace through expansion. Figure 7 (b) supplements an example of a single landlord's evolution in its portfolio size and density. Color gradients show that average costs decline as portfolio scope and density increase, indicated by $\hat{\gamma}_3, \hat{\gamma}_5 < 0$ and $\hat{\gamma}_4 > 0$. From year 0 to year 8, we observe overall increases in both portfolio size and density in the majority of locations, resulting in decreases in maintenance costs per unit.

To quantify the margin of cost reduction from portfolio growth, we conduct a counterfactual analysis using landlords' initial portfolio size and densities to calculate the per-unit costs for units in landlords' post-expansion portfolios in Year 8 (2021). We then compare the real costs with the counterfactual costs at the firm-market level. Figure 7 (d) shows that portfolio growth positively contributes to cost reduction. The aggregated cost reduction rate across all firms and primary markets is 60.03%, or \$1312.5 per rental unit month in 2021.

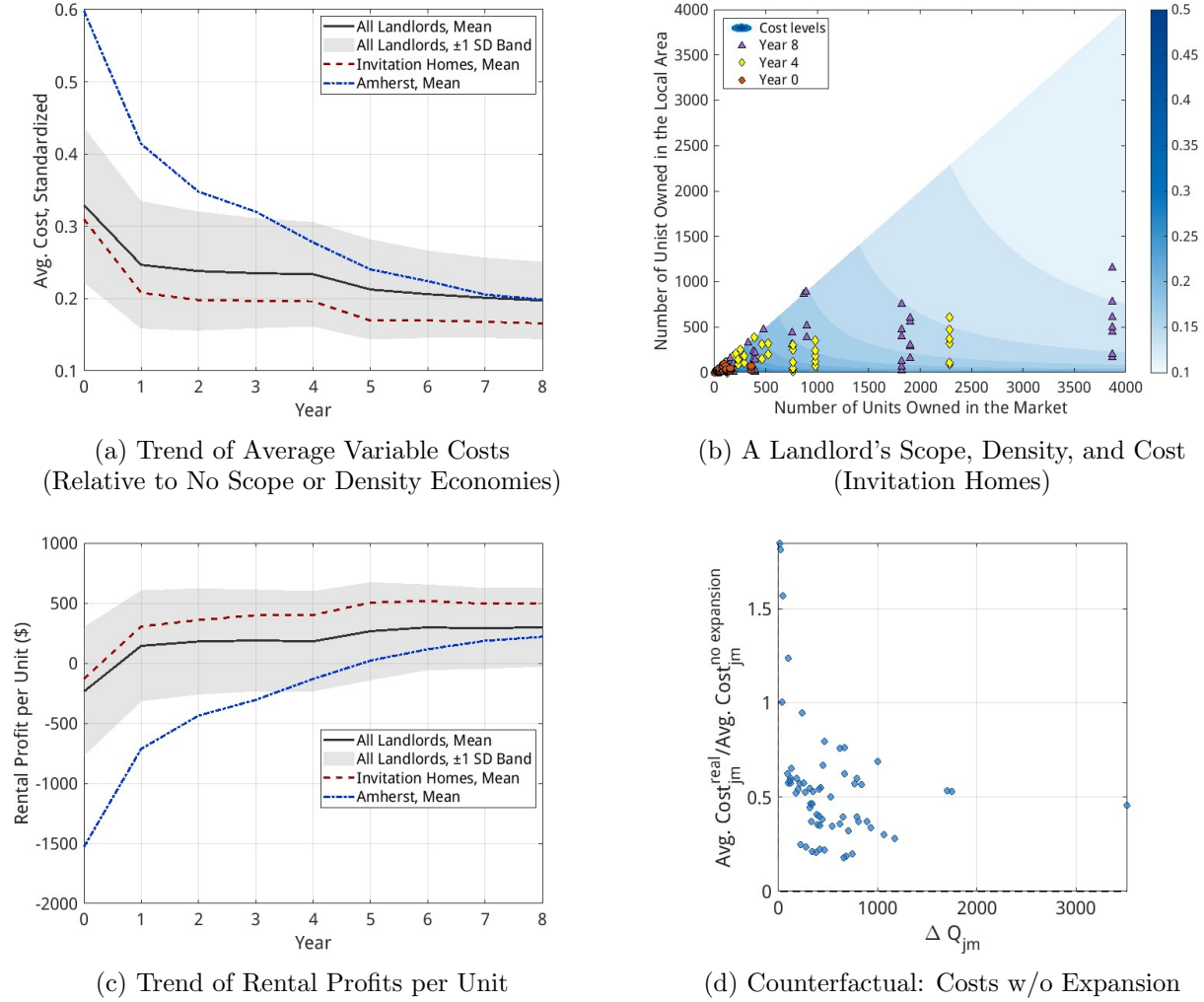


Figure 7: Firm Dynamics in Scope, Density, Cost, and Rental Profit

Notes: In Panel (a) and (c), we plot the trend of average variable costs and per-unit rental profits, for rental homes owned by any of the institutional landlords (black solid), Invitation Homes (red, dashed), and Amherst (blue, dot-dashed). For the industry trends we provide a bandwidth of one standard deviation, visualizing the dispersion of variables of interest across institutional rental units in a given year. In Panel (b), we visualize the within-portfolio evolution of a single landlord's scope and density on a color gradient, representing a continuum of average variable cost levels, relative to the cost level of operating one isolated unit with no scope or density economies. We plot snapshots of three years, Year 0 (red), Year 4 (yellow), and Year 8 (purple), for each the scattered points represent the values of $(Q^{total}, Q^{local})_{hjmt}$ across h in landlord j 's stock. In Panel (d), we compare portfolio-average costs at firm-market level w/o Expansion since year 0, and plot changes in fraction against expanded portfolio size from year 0 to year 8.

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Supplement to “The Expansion and Dynamic Equilibrium Effects of Institutional Landlords”

A Notation

Table 10: Description of Variables

Symbol	Description
m	index of geographic market
t	index of year
h	index of house type
j	index of institutional landlord
i	index of household
d	index of demographic type
λ	index of home owner type, $\lambda \in \{oo, sl, il, new\}$
$\mathcal{H}_m$	set of house types in market m
$\mathcal{J}_{mt}$	set of active institutional landlords in market m year t
$P_{D,mt}$	demographic distribution of housing population in market m year t
Q_{hmt}^λ	number of units owned by owner type λ
Q_{jhmt}	number of units owned by branded landlord j
$\xi_{hmt} = (\xi^O, \xi^{R,sl}, \xi^{R,il})_{hmt}$	product level demand shocks
$\epsilon_{ihmt} = (\epsilon^O, \epsilon^{R,sl}, \epsilon^{R,il})_{ihmt}$	idiosyncratic household-product-market-year demand shocks
Φ_{jmt}	flow payoff for landlord j in market mt
avc_{hjmt}	average variable cost for landlord j to operate rentals in house type h
ω_{mt}	real estate employee wage in market m year t
c_{jmt}^I	investment cost for landlord j in market m year t
ν_{hjmt}	house type-landlord specific per-unit investment cost shock
$\mathcal{M}$	common state variables

B Variable Construction

B.1 House characteristics

We use a combination of houses’ physical attributes and neighborhood amenities to characterize houses and categorize them into differentiated types.

Bedroom type: We categorize houses into four bedroom types with ≤ 2 bedrooms, 3 bedrooms, 4 bedrooms, and ≥ 5 bedrooms. For properties with missing data in `#bedrooms` in all of its tax records, we predict their number of bedrooms using the distribution of `#bedrooms` from the observed set of houses built on the same land parcel. [I may include some evidence to show the quality of prediction]

Neighborhood walk score: We web-scrape neighborhood walk score from Zillow²⁶, using a

²⁶Zillow’s walk score is sourced from Walk Score, a company that provides walkability ratings for neighborhoods

representative sample of SFH properties located across Atlanta neighborhoods. We use the mean value of sampled properties' walk score within a tract to form the tract-level walk score.

Indicators for access to high-quality schools: We web-scrape nearby school information from Zillow²⁷ using the same sample of properties as described above. We summarize the raw information on a house's nearby schools - school name, category, and scores (from 1 to 10) - into three binary variables: indicator for access to high-quality²⁸ preschools or elementary schools, indicator for access to high-quality middle schools, and indicator for access to high-quality high schools.

B.2 Monthly cost of homeownership

We calculate the monthly cost of home ownership (p_{ht}) as the sum of mortgage principal and interest payment, property tax payment, and home insurance²⁹, that a representative home buyer is expected to pay every month if buying house h in county m in year t :

$$p_{hmt}^O = p_{hmt}^{O, \text{lump sum}} \cdot ((1 - d) [r^{\text{mtg}}(1 + r^{\text{mtg}})^N] / [(1 + r^{\text{mtg}})^N - 1] + r_{mt}^{\text{tax}}/12 + r_{hmt}^{\text{insurance}})$$

- Benchmark mortgage: 30-year fixed-rate mortgage with 20% down payment. We collect mortgage rates each year using the level during first week in July
- Property tax rate: the median of empirical tax rates (ratio between tax amount and property's market value) calculated from all SFH properties within the same county and year
- Home insurance premium rate: the median of empirical home insurance rates (ratio between estimated home insurance by Redfin and property sale price) calculated from all SFH properties listed in the MSA. In the Atlanta MSA sample, Redfin data shows little variation in premium rate across SFH properties.

We exclude property transactions with missing prices, inter-family transactions, and multi-property transactions with missing land size in any property transacted, from constructing type-level p_{hmt}^{sale} . For multi-property transactions with all properties' land sizes documented, we allocate the total sale amount to each property based on its proportionate land size.

based on factors including the proximity to amenities, the density of intersections, and pedestrian-friendly features. Website: <https://www.walkscore.com/>

²⁷Zillow identifies nearby schools for each property by combining local school district boundary data with the property's location. School score is sourced from GreatSchools, a website providing school ratings and comparison tools based on student growth, college readiness, equity, and test scores for public schools in the U.S. Website: <https://www.greatschools.org/>

²⁸A school is defined as of high quality if its score exceeds the median of all Atlanta schools in the same grade level.

²⁹Home insurance is mandated by most lenders if the home is bought with a mortgage.

Table 11: Summary of Institutional Landlords and Subsidiaries Cleaning

Brand Name	Code	Years in Operation	Acquired by	Type	No. Subsidiaries Identified
HavenBrook Partners	HB	2012 - 2018	FYR	Private	7
Front Yard Residential	FYR	2012 - 2020	PR	Public	35
Progress Residential	PR	2012 - present	-	Private	132
Starwood Waypoint Homes	SFR	2012 - 2017	INVH	Public	107
Invitation Homes	INVH	2012 - present	-	Public	67
Silver Bay Realty	SBY	2012 - 2017	TAH	Public	15
Tricon American Homes	TAH	2012 - present	-	Public	64
First Key Homes	CCM	2015 - present	-	Private	59
American Residential Properties	ARPI	2008 - 2015	AMH	Private	7
American Homes 4 Rent	AMH	2012 - present	-	Public	69
Amherst Residential	AMHST	2012 - present	-	Private	155

Notes: We only include subsidiaries that participated in Atlanta MSA's single-family house market. Subsidiaries added via M&As are not counted into the acquirer's No. Subsidiaries directly. The same subsidiary can be counted multiple times if the spelling of its name as a buyer is inconsistent across transaction records.

B.3 Institutional ownership

Table 11 summarizes the corporate ownership cleaning results for institutional landlords in Atlanta MSA.

C Tables and Figures

Table 12: First Stage Regressions: IVs for Static Supply Estimation

	$\log(p^{O, \text{lump sum}})$	$\log(p^{O, \text{lump sum}}/p^{R-sl})$
(Intercept)	5.901 (0.221)	-2.110 (0.228)
share_income_50to100k	1.518 (0.076)	0.465 (0.078)
share_income_100to150k	2.230 (0.119)	0.853 (0.124)
share_income_150to200k	3.323 (0.222)	0.614 (0.230)
share_income_200kup	2.940 (0.216)	1.082 (0.226)
share_race_white	-1.737 (0.167)	-0.185 (0.173)
share_race_black	-1.129 (0.180)	-0.284 (0.186)
share_race_asian	-1.134 (0.280)	-0.100 (0.290)
share_race_hispanic	-0.569 (0.223)	-0.020 (0.232)
share_educ_bachelor	0.203 (0.082)	-0.187 (0.085)
share_age_25to35	-0.885 (0.181)	-0.000 (0.188)
share_age_35to45	-1.506 (0.190)	-0.234 (0.198)
share_age_45to55	-0.765 (0.197)	-0.112 (0.205)
share_age_55up	0.020 (0.187)	0.010 (0.195)
FEs: home type	X	X
Multiple R^2	0.886	0.640
Adjusted R^2	0.870	0.591
F-statistic	58.85	13.04
Obs.	6,899	5,726

Notes: This table summarizes the first-stage regression results for demographics instruments used in estimation of static supply functions. We measure the total ownership price of houses, $p^{O, \text{lump sum}}$ in \$1,000, not including property taxes, mortgage interests, or insurances. We measure the rental prices of small landlords, p^{R-sl} using the monthly rents in \$.

Table 13: Frequency of Institutional Landlords' Net Purchases (per Year) in House Types

Institution	Types ever invested									Never invested
	<-5	-5 - -1	0	1-5	6-10	11-15	16-20	21-25	>25	0
Invitation Homes	52	373	2521	1109	263	105	57	41	114	2493
Progress Residential	0	60	1958	1111	241	93	55	24	67	3519
American Homes 4 Rent	4	182	1583	736	128	52	17	6	10	4410
Amherst	2	33	1861	999	164	68	32	23	58	3888
Tricon American Homes	5	71	1796	629	122	42	23	13	26	4401
FirstKey Homes	0	18	2476	928	183	54	37	19	29	3384
Starwood Waypoint Homes	16	308	1122	450	118	53	33	11	44	1805
Front Yard Residential	2	67	1711	512	69	28	14	10	27	3896
HavenBrook	46	171	600	247	41	11	2	3	7	3624
Silver Bay	4	104	685	359	48	16	10	0	9	2725
American Residential Properties	1	11	401	201	33	13	3	2	4	1707

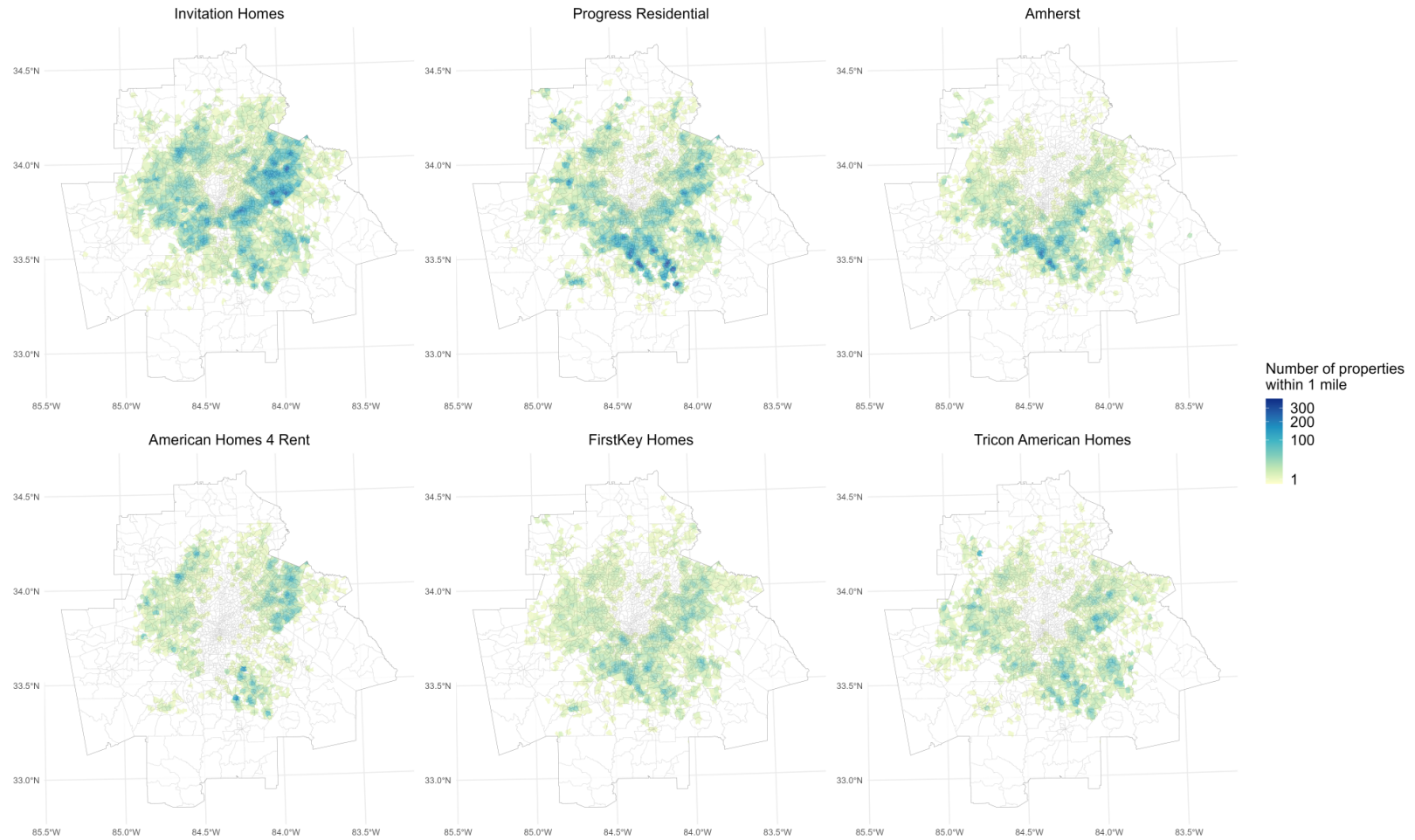
Notes: ...

Table 14: Correlation between Demographics and Housing Characteristics

	Monthly cost (\$1,000)	Three- bedroom	Four- bedroom	Five- bedroom	Indicator: ownership	Indicator: SFH
log(income)	0.325	-0.062	0.116	0.142	—	0.142
Income group: \$50k-\$100k	—	—	—	—	-0.039	—
Income group: \$100k-\$150k	—	—	—	—	0.147	—
Income group: \$150k-\$200k	—	—	—	—	0.175	—
Income group: \$200k+	—	—	—	—	0.258	—
Number of child: 1-2	0.062	0.019	0.072	0.057	0.058	0.196
Number of child: > 2	0.016	-0.060	0.106	0.113	-0.057	0.143
Age: 25-35	-0.023	—	—	—	-0.013	-0.103
Age: 35-45	0.051	—	—	—	-0.051	0.080
Age: 45-55	0.047	—	—	—	0.029	0.098
Age: 55+	-0.056	—	—	—	0.066	0.018
Bachelor's Degree	0.389	—	—	—	0.277	—
Race: White, non Hispanic	0.124	—	—	—	0.142	—
Race: Black	-0.174	—	—	—	-0.156	—
Race: White, Hispanic	-0.037	—	—	—	-0.055	—
Race: Asian	0.114	—	—	—	0.089	—

Notes:

Figure 8: Rental Portfolio Density of Six Institutional Lanldords, Atlanta 2022



Notes: We measure the local density of a landlord's rental portfolio using the number of SFH units operated by this landlord located within 1 mile of the point of evaluation. We visualize the densities evaluated at census block centroids. By 2022, all six institutional landlords have achieved substantial levels of portfolio density with differentiated target locations.